Advancing K-14 Invention Education Pathways: Insights from Regional Partners and Collaborators

Prepared by

LMIT Student Fellows:

K'vion Berry

Alexa Bohanon

Believer Gall Hirwa

Caroline Laber

Parker McClain

Sophie Thompson

LMIT Personnel:

Dr. Tamara Galoyan, Research, Evaluation, Assessment Specialist

Dr. Stephanie Couch, Executive Director

Dr. Michael Cima, Faculty Director

August 2026

Lemelson-MIT Program

Massachusetts Institute of Technology

Table of Contents

Executive Summary 2

Introduction to the Study 4

Student Research Fellows 2026 Program 5

Research Design 7

Problem and Research Questions 7

Study Design 8

Participant Recruitment 9

Data Collection 9

Data Analysis 10

Participants 11

Findings 12

Overall Findings for Conceptualizations of IvE 16

Key Trends for Conceptualizations of IvE by Region and Grade 20

Overall Findings for Capacity 24

Key Trends for Capacity by Region, Role, and Grade Level 36

Overall Findings for Barriers 41

Key Trends for Barriers by Region, Role, & Grade Level 52

Overall Findings for Partner Needs 56

Key Trends for Partner Needs by Role, Region & Grade Level 58

Overall Findings for Future Goals…… 62

Key Trends for Future Goals by Region, Role, and Grade Level 64

Discussion 68

RQ1. How do regional partners conceptualize and implement Invention Education? 68

RQ2. What conditions influence the development and scaling of K–14 IvE pathways? 69

RQ3. What role should LMIT play in supporting these efforts? 70

Conclusion 71

References 72

Appendix: Interview Guide 73

Executive Summary

The Lemelson-MIT (LMIT) program seeks to expand access to high-quality Invention Education (IvE) by developing connected PreK-14 pathways that integrate invention experiences into formal education and strengthen connections among schools, districts, higher education, industry, nonprofits and community organizations. Our focus on Prek-14 pathways emerged from our awareness that student success with invention-oriented projects is enabled by many different types of experiences and opportunities for learning across time. The connection to supporters in the local ecosystem recognizes that educators need ongoing support (including filling gaps in their own knowledge or skills required to support their learners' projects). LMIT conducted a qualitative interview study to gain baseline understanding and provide empirical evidence of educators' perspectives and beliefs about conditions that they believe will support or constrain the implementation and scaling of IvE. The study was conducted as part of LMIT's annual Student Research Fellows' program between June and August 2026. Six MIT student fellows, whose day-to-day work was led by Dr. Galoyan, conducted semi-structured interviews with 25 partners in three geographical regions (Northern California, Tennessee, and Massachusetts / Connecticut) with whom LMIT will work over the next three years (2027-2029). Study participants included teachers (40%), school or district administrators (32%), industry partners (12%), and other community or educational stakeholders (e.g., directors of a center or a lab) (16%). The study focused on three partner regions including Northern California, Tennessee, and Massachusetts. Researchers examined three topics: 1) how regional partners conceptualize and currently implement IvE; 2) what capacities, barriers, supports, and partnerships influence pathway development and scaling; and 3) what role LMIT can play in supporting these efforts. Findings were reported in five broad domains that emerged during the course of the study: *Conceptualizations of IvE, Capacity, Barriers, Partner Needs,* and *Partners' Future Goals*.

Conceptualizations of IvE. Across regions, participants viewed IvE as a flexible, authentic approach that connects learning to real-world problems while developing durable and technical skills. They also emphasized its potential to strengthen students' agency and identities as problem solvers, innovators, and inventors.

Capacity. Partners reported existing curricula, courses, extracurricular and CTE programs, professional development, resources, assessments, and academic and industry partnerships that can support IvE pathway development. However, capacity varied across

contexts, and many existing IvE related opportunities remain individual experiences rather than intentionally connected K-14 pathways.

Barriers. Key challenges the partners identified as they set out to create K-14 IvE pathways include limited time, competing school priorities, funding and resource constraints, staffing and teacher capacity, and difficulties integrating IvE into existing curricula and assessment systems. Participants also identified challenges in creating continuity across grade levels and providing educators with sufficient flexibility to implement IvE.

Partner Needs. Partners emphasized the need for teacher training, funding, administrative support, adaptable curriculum and assessment resources, and stronger collaboration. They also sought better mechanisms for identifying regional partners, communicating IvE to stakeholders, and sharing knowledge within and across regions.

Partners' Future Goals. Partners sought to strengthen existing IvE curricula and assessment while expanding opportunities across schools, communities, and grade levels. They emphasized developing more intentional K-14 pathways that establish shared outcomes while remaining flexible enough to reflect regional priorities, resources, and contexts.

Across the five domains, a central finding is that the three regions are not starting from scratch. Each has existing programs, expertise, partnerships, and resources that can provide a foundation for pathway development. At the same time, these assets are distributed differently across regions and organizations and are not yet consistently connected into coherent K-14 pathways. The findings therefore suggest that scaling IvE will require moving from individual programs and opportunities toward coordinated regional ecosystems that connect educators, schools, higher education, industry, and community organizations while allowing pathways to remain responsive to local contexts and priorities.

Introduction to the Study

Preparing learners to address complex, ill-structured societal challenges such as sustainability, public health, technological innovation, and social equity requires educational approaches that extend beyond the boundaries of singular disciplines. Invention Education (IvE), a transdisciplinary pedagogical approach rooted in project and problem-based learning (PBL), cultivates the creative problem-solving, technical and human durable skills (such as teamwork and collaboration) needed to navigate the complex challenges people face in real-world contexts. Students engaged in IvE offerings identify meaningful community problems, root causes, and unmet human needs. Students collaboratively design, iterate, and refine solutions that create social value (Dalela & Ahmed, 2024). This emerging pedagogical approach integrates processes and practices from engineering design, scientific inquiry, computational thinking, social sciences and humanities, entrepreneurship, and community engagement, in ways that expand traditional problem-based learning to include authentic invention experiences grounded in learners' interests and local contexts (Couch et al., 2019).

Despite the benefits of IvE, invention-oriented learning primarily occurs in informal and afterschool STEM settings such as summer camps, interactive museum exhibitions, maker spaces, and libraries (Lemelson Foundation & Coy, 2020; Garner et al., 2021), which restricts access across cultural, linguistic, racial, and socioeconomic groups (National Academies of Sciences, Engineering, and Medicine, 2024a). Researchers call for actions to “democratize IvE education and empower educators to contextualize opportunities within diverse learning environments” (Dalela & Ahmed, 2024, p. 15). Federal policymakers also call for expanded access to IvE in their STEM education plan for the United States (National Academies of Sciences, Engineering, and Medicine, 2024b).

For more than three decades, the Lemelson-MIT (LMIT) Program, in partnership with The Lemelson Foundation, has advanced the vision that every learner can become an inventor when provided with meaningful opportunities to identify and solve authentic problems that improve lives and strengthen communities. Through research, educator professional development, curriculum development, student programs, and strategic partnerships, LMIT has helped establish IvE as an evidence-informed educational approach that develops both technical competencies and durable human skills. Building on this foundation, LMIT's current strategic vision is to make high-quality IvE accessible throughout the PreK-14 educational pathway by

embedding invention experiences within the school day and strengthening connections among academic disciplines, career and technical education (CTE), higher education, industry, and community organizations.

Achieving this vision requires more than effective instructional materials or professional learning opportunities. Sustainable implementation depends on the development of regional ecosystems in which educators, schools, colleges and universities, industry partners, nonprofit organizations, community leaders, and policymakers work collaboratively to create coherent pathways for IvE. These partnerships help expand access to authentic IvE experiences, align educational opportunities across grade levels, strengthen workforce connections, and build the infrastructure necessary for long-term implementation and sustainability. By cultivating these regional ecosystems, LMIT seeks to establish scalable and equitable PreK-14 pathways that enable learners to develop the knowledge, skills, and mindsets needed to become creative problem solvers, innovators, and community leaders. Some may even generate intellectual property that aids business start-ups and create jobs needed for gainful employment.

To better understand the conditions necessary to build and sustain these regional ecosystems, the LMIT Program conducted a qualitative study in the summer of 2026 across three partner regions: Northern California, Tennessee, and Massachusetts/Connecticut (for programmatic purposes, Massachusetts and Connecticut are included within the same region of interest by LMIT). Semi-structured interviews were conducted with 25 educators, school and district leaders, and industry representatives engaged in advancing IvE. The purpose of the study was to examine stakeholders' perspectives on regional capacity, partnerships, opportunities, challenges, and support needs related to implementing and sustaining Invention Education pathways. The findings presented in this report are intended to inform LMIT's work with the three regions, identify opportunities for strengthening regional collaborations within and across regions, and provide empirical evidence for expanding access to high-quality IvE across diverse educational contexts.

Student Research Fellows 2026 Program

The MIT Student Fellows Program was launched in the summer of 2022 to assist LMIT with studies of IvE offerings and their impact using qualitative and ethnographic approaches. Ethnography is used in academic research to study a wide range of cultural phenomena as well as

in the private sector to understand consumer needs or user experiences with new products. Integrating ethnographic approaches into studies of LMIT programs and offering opportunities for MIT students to learn ethnographic perspectives by engaging in research enables MIT students in many different disciplines to explore the intersections of social science methodologies, education, and their fields. The year 2026 marks the fifth year of the program's six-week summer research internship. This year, the internship was facilitated by Dr. Tamara Galoyan. LMIT recruited six fellows who participated in an intensive study of qualitative and ethnographic research approaches focusing on interviewing, interview data analysis, and reporting findings.

Student Fellows

Background information for the six participating 2026 Student Fellows whose contributions are consequential for this report are as follows:

K'Vion Berry is a class of 2029 undergraduate at MIT who's majoring in Electrical Engineering and Computer Science. He also works on mechanical engineering research at MIT under the Lincoln Laboratory. Some things he enjoys doing include playing soccer and tennis as well as online FPS games.

Alexa Bohanon is a class of 2029 MIT undergraduate studying Computer Science and Artificial Intelligence with a minor in music. She has previously conducted probabilistic programming research at MIT's Computer Science and Artificial Intelligence Laboratory (CSAIL) and served on the software subteam of the Human Technology Integration Club. Beyond academics, she plays guitar and sings with MIT LIVE and competes in javelin for MIT's varsity track and field team.

Believer Gall Hirwa is an MIT Class of 2029 undergraduate studying Electrical Engineering and Computer Science. Interested in using engineering and creative problem-solving to understand complex challenges and develop practical solutions.He works at the intersection of computing, hardware, and engineering, which he explores through materials research in MIT's Department of Materials Science and Engineering (DMSE). Outside the classroom and lab, he enjoys practicing Shotokan and Wadō-ryū Karate.

Caroline Laber is a rising sophomore at MIT studying Chemical-Biological Engineering and a member of the MIT Women's Swimming and Diving Team. During my first year, she

conducted research focused on developing more sustainable methods for producing materials, an experience that strengthened my passion for using engineering to address real-world challenges. Caroline is particularly interested in biomaterials and medical research, with the goal of advancing technologies that improve health and quality of life.

Parker McClain is a rising Junior studying Chemical Engineering at MIT. He is a member of the varsity football team and a TA for Physics 1, Physics 2, and Organic Chemistry. He loves playing board games, sailing, skiing, music, reading, good coffee (or bread), being outside, and trying new things. Most recently, Parker worked in the Plata lab looking at biodegradable plastics. He is currently working on a startup that helps to improve health outcomes for people with type 1 diabetes through predictive algorithms.

Sophie Thompson is a recent MIT graduate in Chemical Engineering whose research interests lie at the intersection of human centered design and healthcare. At MIT, she conducted research in the Biomechatronics Group, focusing on optimizing the manufacturing method of a banana fiber prosthetic socket for use in low-resource clinical environments. She also held leadership positions in TEDxMIT and AIChE, two organizations dedicated to bridging the gap between scientific innovation and the public. Through the Lemelson-MIT program, she is excited to contribute to the expanded access to STEM education.

Facilitator

Dr. Tamara Galoyan is a Research, Evaluation, and Assessment Specialist at the Lemelson-MIT Program. Tamara's research is interdisciplinary and is at the intersection of Learning Sciences, STEM education, and learning technologies. She has more than a decade of professional experience in educational research, teaching graduate and undergraduate courses, serving on dissertation committees, curriculum development, program evaluation and assessment, and teacher training and mentoring in the context of PreK-12, higher education, and continuous workplace learning.

Research Design

Problem and Research Questions

While researchers and policymakers increasingly advocate for expanding IvE within formal education, limited research has examined the organizational and regional ecosystem

conditions necessary to support its implementation, growth beyond initial adopters, and long-term sustainability. Understanding these conditions is essential for developing equitable, scalable PreK-14 IvE pathways that prepare all learners to address complex societal challenges.

The study was guided by the following research questions:

RQ1: How do LMIT's regional partners currently conceptualize and implement Invention Education (IvE) within their organizations and regions?

RQ2: What conditions (capacities, barriers, supports, and partnerships) influence the development and scaling of K-14 IvE pathways?

RQ3: What role should LMIT play in supporting partners and regional efforts?

Study Design

This study employed qualitative methodology with 25 semi-structured interviews. The study was informed by an ethnographic perspective emphasizing the co-construction of meaning between researchers and participants (Brinkman & Kvale, 2015; Roulston, 2022). Interviews sought to understand participants in their own language and context, with responses shaped by interactional styles, rapport, timing, and question framing. Qualitative research values the uniqueness of each participant's perspective while also identifying patterns across cases, aiming for transferability rather than statistical generalizability. Trustworthiness was fostered through transparency, explicit evidence, and reflexivity (Davies & Dodd, 2002; Tracy, 2010). The research process was highly collaborative, where the fellows and facilitator met regularly to align research practices, ensure coding consistency, and develop claims grounded in data. Daily reflections and discussions among the researchers deepened reflexivity, supporting the credibility and applicability of findings.

The study received approval from MIT's institutional review board, known as the Committee of the Use of Humans as Experimental Subjects (COUHES), before the researchers started recruiting participants and collecting data. As part of this process, all student fellows completed the CITI (Collaborative Institutional Training Initiative) training and certification needed for behavioral and social science research.

Participant Recruitment

Participants for this study were recruited during Summer 2026 using two primary methods. We first recruited participants during LMIT's Regional Partners' Convening in June 2026, where Dr. Galoyan introduced the research study to attendees. During this session, Dr. Galoyan provided an overview of the study's purpose, goals, and procedures, explained what participation would involve, and invited eligible regional partners to participate in individual interviews by completing a virtual sign-up sheet. Attendees were also informed that participation was entirely voluntary and that their decision to participate or decline would not affect their relationship with LMIT or any ongoing partnership activities.

The second recruitment strategy involved a follow-up invitation email sent to regional partners by LMIT's two regional managers, Dr. Cristina Saenz and Edwin Marrero. The email described the purpose of the study, the types of participants being sought, the estimated interview duration, and the voluntary nature of participation. Interested individuals were invited to respond to the email and indicate their availability for an interview during July 2026. Follow-up reminders were sent as needed to maximize participation across the three participating regions (Northern California, Tennessee, and Massachusetts/Connecticut).

Data Collection

The main instrument for data collection was a semi-structured interview guide designed by Dr. Galoyan and Dr. Couch and reviewed by Dr. Cima (see Appendix A). The interviews were conducted, recorded, and transcribed via Zoom and lasted approximately 45 minutes. The interview guide consisted of several broad opening questions and related prompting questions focusing on participants'current capacity, barriers, and needs in terms of implementing and scaling IvE within their organizations and across regions. In total, the fellows conducted 25 interviews across 11 days. Each interview was conducted by two fellows. The fellows started each interview by introducing themselves and obtaining verbal consent from the participant.

All the interviews were accompanied by note-taking by the interviewers. The notes allowed the interviewers to tailor the follow-up questions to individual participant responses. The notes were also shared with the group in the discussion sessions, which allowed the interviewers to reflect on and discuss their interviewing experiences with the other group members and the facilitator.

Data Analysis

The dataset for this study consisted of 25 de-identified interview transcripts collected from regional partners across Northern California, Tennessee, and Massachusetts. Prior to analysis, all transcripts were de-identified and assigned a unique participant identifier (e.g., Participant 1, Participant 2). A separate master key linking participant names to identification numbers was maintained in a password-protected file accessible only to the researchers of this study.

Interview data were analyzed using an abductive analysis approach. The analysis began with the development of five broad domains informed by the study's research questions and interview protocol: (1) Conceptualizations of Invention Education (IvE), (2) Capacity, (3) Barriers, (4) Needs, and (5) Future Goals. These broad domains served as an initial deductive coding framework to organize the data while remaining flexible to capture new findings.

During the first cycle of analysis, fellows independently generated preliminary categories and subcategories inductively from participants' responses. As additional transcripts were analyzed, the research team continuously refined the codebook by applying existing codes, revising code definitions, merging overlapping codes, and reorganizing the coding structure to better reflect the data. This iterative process allowed the coding framework to evolve while maintaining alignment with the study's research questions.

To support the rigor and trustworthiness of the analysis, each transcript was independently coded by two research fellows. After completing their independent analyses, fellows met to compare interpretations and reconcile differences through discussion and consensus. During these meetings, fellows reviewed the generated codes and coded excerpts and collaboratively determined whether codes should be merged, divided into more specific categories, reassigned to different categories, or applied differently to better represent participants' intended meanings. When consensus could not be reached immediately, questions were brought to the larger research team for discussion. To further support consistency across the dataset, the fellows participated in regular whole-group analytic meetings facilitated by the lead researcher. During these sessions, the team reviewed newly emerging codes, refined definitions, discussed challenging excerpts, and established shared decision-making procedures regarding code application.

All qualitative data management and coding were conducted using Dedoose, a web-based qualitative and mixed-methods data analysis software designed to facilitate the organization, coding, retrieval, and comparison of qualitative data across multiple researchers. Dedoose enabled the research team to collaboratively code transcripts, manage an evolving codebook, retrieve coded excerpts, and examine and visualize patterns within and across participants and regions, thereby supporting systematic and transparent analysis.

The analysis also involved visualizing the findings using the *Descriptor x Code data visualization* feature in Dedoose. This feature allows researchers to visualize the frequency of code application within a descriptor category. In other words, the chart displays the number of coded excerpts disaggregated by subgroups. In our case, the descriptor subgroups included participants' *Region (CA, MA, TN), Grade Level (Elementary, Middle, High, School District, University or College), Primary Role (Teacher, Administrator (i.e., School Principal, District Administrator), Industry Partner,* and *Other (i.e., individuals who held other positions, including program or lab directors, etc.).* All the exported charts in this report represent the normalized percentages of code application. The normalization function allows for a direct comparison between descriptors, accounting for an unequal number of cases within each category.

Several strategies were employed to enhance the trustworthiness of the findings. First, triangulation was achieved through independent coding by multiple researchers followed by consensus-based reconciliation. Second, the iterative refinement of the codebook and regular team discussions established transparency of the process. Third, analytic memos, team discussions, and daily reflection posts shared in Canvas encouraged reflexivity by prompting fellows to examine their assumptions and interpretations throughout the analytic process. Finally, rich descriptions and representative participant quotes were captured throughout the analysis to ensure that the reported findings remained grounded in participants' perspectives and accurately reflected the breadth and depth of the interview data.

Participants

Our sample included 25 participants representing different geographic regions, organizational roles, and grade levels. Participants were recruited from Northern California (44 %), Massachusetts (36%), and Tennessee (20%). Our sample represents 35% of the total of 72

regional partners that are included in LMIT's partner list as of October, 2026. The broader partner population included 64% from Northern California, 25% from MA, and 11% from TN.

Our sample's professional roles included teachers (40%), school or district administrators (32%), and industry partners (12%), and other community-based or educational roles (16%). Participants also represented multiple educational levels, including high school (28%), elementary school (16%), university or colleges (16%), school district (28%), and middle school (4%). Two of the participants were not affiliated with a specific grade level.

Findings

In alignment with our research questions, we grouped the findings from the qualitative data analysis under five major domains, including Conceptualizations of IvE, Capacity, Barriers, Partner Needs, and Future Goals. The domains were created based on the study's research questions and related questions in the interview guide. Within each domain, we organized the codes into related categories and subcategories, as shown in Table 1 below.

Table 1

Findings Organized by Groups, Categories, and Subcategories.

Domain	Category	Subcategory	Excerpt Count
Conceptualizations of IvE *How do regional partners describe IvE? How do they see IvE? What words/terms do they use?*	Learning Outcomes *Student learning outcomes from participating in IvE*	Soft Skills	117
		Technical Skills	19
		Students Mindset and Identity Shift	30
	Connecting Learning to Real World *IvE helps provide context and application to learning*	Contextualizing Learning	37
		Workforce Preparation	14
		IvE is Adaptable	35
	Teacher Opinions of IvE Effectiveness	IvE Drives Learning	20

	Views from teachers surrounding IvE and its impact both in their lives and on the world	Teacher Opinions of Implementation	36
Capacity *Current resources, supports, partnerships that the partners have, current courses related to IvE*	Current Pathways *Current examples of IvE Pathways*	Building IvE Pathways	10
		K-12 Pathway Examples	19
		CTE Pathways	12
		Region-Dependent Pathways	6
	Teacher Professional Development *Different types of training opportunities for pre-service teachers and teachers*	E-lab Training	4
		Training Methods/Resources	21
		Current professional development capacity	9
	Current Curriculum *Current Curriculum offerings related to IvE. Includes things like class offerings and extracurriculars*	Types of Learning	38
		State-Sponsored Curricula	10
		InventEd	10
		Extracurricular Programs	14
		Class Offerings	20
	Current Partnerships *Partnerships between participant interviewed (and the organization they represent) and other organizations*	Industry Partner	65
		Academic Partnership	33
	Current Assessment *Different types of assessment used in partner organization classrooms to assess students. These assessment can be both for IvE and non-*	End-of-Course Showcase	4
		Traditional Assessment	6
		Reflective & Formative Assessment	43
		Real-World Impact Assessment	3

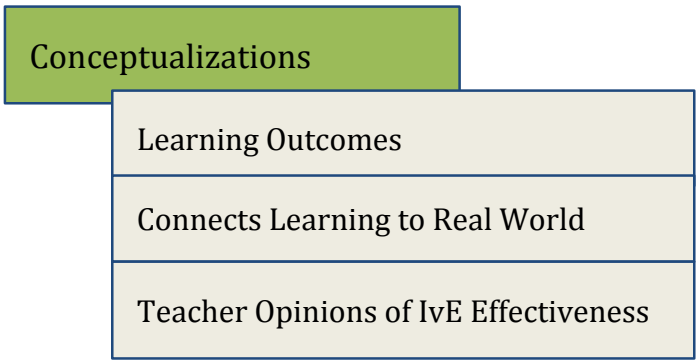
	IvE curriculum		
	Current Resources *Current Curriculum offerings related to IvE. Includes things like class offerings and extracurriculars*	Current Material Resources	21
		Logistical Solutions	13
		Financial Allocations	17
Barriers *Challenges related to IvE implementation*	Lack of Resources *Challenges regarding resource constraint for IvE expansion*	Community Support	6
		Budget/Funding Barriers	21
		Physical Resources	7
	IvE Curriculum *Challenges integrating IvE into school curriculum*	Difficulty Finding Problems	7
		Lack of Assessment	3
		Teacher Confusion Around Assessment	8
	General Education Challenges *Challenges from sources outside of staff/school control*	Worry About AI Uncertainty	7
		Students Favor AP Courses	2
		Phone/Social Media Use	2
	Staffing Barriers *Challenges with implementing IvE due to teacher and administrative staffing issues*	Challenges Obtaining/Convincing Staff	10
		Training-Related Challenges	20
		Teacher Capacities	21
	School Logistical Constraints *Challenges with implementing IvE due to logistical constraints in schools*	Time Constraints	26
		Continuity Across Grade Levels	5
		Conflicting School Priorities	27
Partner Needs *What LMIT's regional partners need to*	Resources *Any resources needed by regional partners*	Funding	14
		Methods to Educate Others on IvE	13
		Updates/Fixes in E-Labs	1

continue offering/expanding IvE	*specific to IvE. Includes all types of resources such as funding, space, time, personnel, etc.*	Administrative Support	2
		Teacher Trainings	27
	Partnership *Partnerships and collaboration opportunities that regional partners believe would help expand IvE.*	Faculty Collaboration	14
		New ways to find partners	3
		Communication with other regions	8
		New ways of Facilitating Communication	7
	Classroom Structure *Any changes involving the day to day proceedings of the IvE classroom. This includes things like curriculum, assessments, and teacher needs.*	Teacher Autonomy	7
		Non-Traditional Assessment Strategies	3
		Curricula	4
Future Goals *What partners want the program to look like*	Improving IvE *Partner goals for improving current IvE offerings in the future*	Potential Assessments	13
		Curricula	22
	Expanding IvE *Partner goals for expanding IvE offerings and reach in the future*	Within a School	62
		Within a Community	18

Overall Findings for Conceptualizations of IvE

Conceptualizations of IvE capture how the regional partners think of and view IvE in its current form. The conceptualizations are captured under three broader categories: *Learning Outcomes, Connected Learning to Real World,* and *Teacher Opinions of IvE Effectiveness* (see Figure 1).

Figure 1. *Code Organization for Future Goals*



Learning Outcomes. Learning Outcomes is a category that captures the intended results and effects of IvE by our partner organizations. Within Learning Outcomes, the participants highlighted soft skills or durable skills that teachers hoped students would develop during IvE programs.

There are many skills that were discussed by our partners. One of the most common skills mentioned was collaboration. Participant 16, a school district administrator from MA, shared:

> *"...the biggest hope I have is that all students will see themselves as being able to collaborate with others in various different fields who have various different interests, and that they all understand that they can contribute to that effort*" [Participant 16, School District Administrator, MA]

Similar views were echoed by many other participants around collaboration, but there were also many partners who highlighted critical thinking, identifying problems, and problem-solving skills as some of the main learning outcomes they hoped would come from IvE.

We didn't only hear about soft skills. There was a lot of feedback around hopes that IvE would also be a vehicle for imparting technical skills to students as well. Partners mentioned that they hoped their students could learn entrepreneurial skills and obtain knowledge of how patents work. Participant 20 stated:

> *"It's more like the inspiration of what it takes to be an inventor. Like, you know, how do you write patents if they get to that point? Like, how we do all that*." [Participant 20, School Administrator, MA]

The last emergent theme of this section was that of students gaining a shift in mindset and identity. Our partners discussed how they hoped IvE programs would influence students and how they thought about themselves. Some hoped students would see themselves more as problem solvers and innovators. Others highlighted how they hoped students would be more empowered to solve problems and make change. Participant 24, a university professor from CA, discusses

how they hope IvE will impact their students by stating:

> *"...becoming an inventor or creating something new and novel in this world is oftentimes so little they can't perceive it, right? And like, so a lot of the time, it's the demystifying the practice of being an inventor"* [Participant 24, University Professor, CA]

Connecting Learning to Real World: Connecting learning to the real world is a category within *Conceptualization*, referring to how participants understand IvE as an approach connecting learning to contexts beyond the classroom. We separated this category into three subcategories: *Contextualizing Learning, Workforce Preparation,* and *Invention Education is Adaptable.* Each of these subcategories represents a distinct aspect of how participants define and describe IvE, including its role in connecting learning to authentic contexts, preparing students for future workforce opportunities, and adapting to different educational environments and learner needs.

The subcategory of *Contextualizing Learning* encompasses instances where participants emphasized that IvE provides meaningful context for learning by connecting academic concepts to authentic experiences. Specifically, the participants saw IvE as an approach that connects learning to real world problems, connects Invention Education with history, as well as described it as a study where students invent things and learning that emphasizes real-world application, creativity, and collaboration over memorization. Participant 5, a Project Manager at a museum in CA, when asked on where they see IvE fitting into their curriculum, responded:

> *"...it's sprinkled throughout, so one of our programs is… one of our school programs that's one of the guided ones is called Horses to Horsepower, and we talk about how the railroad… building of the railroad changed California, and how that happened. So they talked about the actual building of the railroad, how that impacted things, and our… actually, our very last station this year was talking about specific problems that were on the railroads, and then people who created inventions to solve it. So that was our last station, out of five stations on there. So it changes a little bit every year. We've had different versions of that.* [Participant 5, Museum Project Manager, CA]

There was an emphasis on focusing IvE on curricula integration, especially in practice, outside the classroom. Another major topic discussed by the participants was the idea that IvE broadens students' perspectives. One instance of this is Participant 3's excerpt:

> *"...people enjoyed the experience and had a good time learning and living there. But because we are so insular, I feel like Invention Education helps to foster greater creativity, and foster a bigger mindset. Like, it's nice to have a tight-knit community, but it's also nice to understand that you're part of a bigger community of our* world." [Participant 3, High School Teacher, CT]

Across transcripts, participants describe IvE as connecting classroom learning to real-world problems, interdisciplinary experiences, and the invention process itself, enabling students to apply knowledge in meaningful ways while developing creativity, collaboration, and problem-solving skill.

Workforce Preparation captures instances where participants portray IvE as equipping students with the knowledge, experiences, and skills needed for future careers and life beyond school. Within this subcategory, the participants discussed three major topics: *students learn real-world STEM work, students figure out careers,* and *it prepares kids for paths other than college.*

Participants emphasized that IvE allows students to experience forms of work that resemble the actual practices of scientists, engineers, designers, and technical professionals. Participant 12, a university professor from CT, described the importance of showing students that they are already capable of engaging in meaningful professional work:

> *"Trying to show them what the actual day-to-day job of a scientist or an engineer would entail. And to show them that they can be trusted with that work now, right? Like, they kind of still have, some of them, the mentality of, like, 'Oh, I'm just a kid, you know, I'm 15 or 16 years old, I can't do this stuff yet. I'll get to this five years from now, ten years from now*." [Participant 12, University Professor, CT]

Participants also described IvE as helping students recognize their interests and make informed decisions about future careers. Participant 23, a high school teacher from TN, explained:

> *"The goal is for students to be able to better know themselves and their strengths and identify a career that is the best fit for them. So we kind of have this saying in our office where we are helping students to make their career by choice, not by chance. And so this is empowering them with the knowledge that they need in order to make an educated decision about a career that's the best fit."*[Participant 23, High School Teacher, TN]

Finally, participants challenged the assumption that college should be the only valued pathway after high school. Participant 19, a school district administrator from TN, reflected on the need to restore career and technical opportunities:

> *"Back in 2001, when No Child Left Behind came into play, it was the push that, hold on, every kid should be able to go to college, right? And so there were a lot of career and technical education opportunities prior to that, and a lot of those went wayward. From 2001 to about 2012, 2015, people were like, no…we have to make school about everybody going to college. We're in 2026, and people are like, we swung the pendulum way too far. We've got to have opportunities to really find out what kids are passionate about, and what*

skill sets we can develop." [Participant 19, School District Administrator, TN]

These excerpts suggest that participants do not view workforce preparation as narrow occupational training. Instead, IvE gives students opportunities to participate in authentic work, understand their strengths and interests, and consider several possible futures.

Invention Education is Adaptable is a subcategory that consists of excerpts and codes where participants characterize IvE as a flexible approach that is modified to address diverse educational contexts, student interest, and societal needs. Within this subcategory, the participants discussed several major topics, including *designing curricula around student interest; IvE shifts courses to be more interdisciplinary and complementary to existing curricula; IvE can foster AI shift*; and others.

Participants frequently described student interest as the starting point for meaningful invention-based learning. Participant 6, a high school teacher from MA, explained:

> *"We could do something within the school building, community; we can take on global issues, but I try to start out first with the student and then work them outward*." [Participant 6, High School Teacher, MA]

Adaptability also appeared in participants' descriptions of IvE as crossing conventional disciplinary boundaries. Participant 11, a university professor from CA, described one of its primary objectives as:

> *"…understanding, like, interconnectedness of different subject matters, so really thinking about interdisciplinary, transdisciplinary approaches to problem*-solving." [Participant 11, University Professor from CA]

Participants further emphasized that flexibility does not mean the absence of structure. Instead, IvE can maintain clear educational goals while leaving space for teachers and students to determine the direction of projects. Participant 4, a school district administrator, summarized this balance:

> *"So it's structured, but it's open-ended. And that's the piece that's really critical, is that open-ended piece."*[Participant 4, School District Administrator, MA]

These excerpts illuminate three related dimensions of adaptability. Participant 6 emphasizes responsiveness to students, Participant 11 emphasizes movement across disciplinary boundaries, and Participant 4 emphasizes curricular openness within an existing structure. They suggest that IvE is adaptable because it provides a common process for invention without requiring every

teacher, student, or school to follow an identical project.

Teacher opinions of IvE Effectiveness. Teacher opinions of IvE effectiveness is a category within *Conceptualization* that encompasses excerpts in which participants express their perspectives on the effectiveness of IvE, including its influence on student learning, teacher development, educational outcomes, and its implementation in schools. It comprises two subcategories: *IvE Drives Learning* and *Teacher Opinions of Implementation.*

The subcategory of *IvE Drives Learning* encompasses excerpts in which participants describe IvE as a driving force for learning and educational improvement. It includes perspectives on how IvE addresses limitations of traditional education, promotes teacher and student learning, strengthens learning literacy, contributes to broader societal progress, and enhances academic achievement. It includes five topics*: IvE fixes gaps in traditional education systems; IvE drives human progress; IvE is a catalyst for learning literacy; IvE drives teacher learning;* and *IvE increases state test scores.*

Teacher Opinions of Implementation encompasses excerpts in which participants discuss their perspectives on the implementation of IvE in schools. For instance, Participant 15, an elementary school teacher from CA, shared:

> *"It requires students to think in a way that's like a computer or AI is not going to replace this type of thinking that students have to do. I think that invention education also really builds empathy.*" [Participant 15, Elementary School Teacher, CA]

Key Trends for Conceptualizations of IvE by Region and Grade

Further analysis of the codes grouped under Conceptualizations of IvE included visualizing findings by descriptor subgroups, including Region and Grade Level (see the Data Analysis section of this report for details on the data analysis and visualization process).

The percentages presented in the figures represent the normalized distribution of coded excerpts across descriptor groups. Overall, the findings indicate both considerable consistency in how IvE was conceptualized across the three regions and meaningful differences in the emphasis placed on particular outcomes and characteristics of IvE.

The development of technical skills was an important dimension of how participants conceptualized IvE, although the distribution of these mentions varied across grade levels. Middle school accounted for the largest proportion of technical-skills references (29.2%).

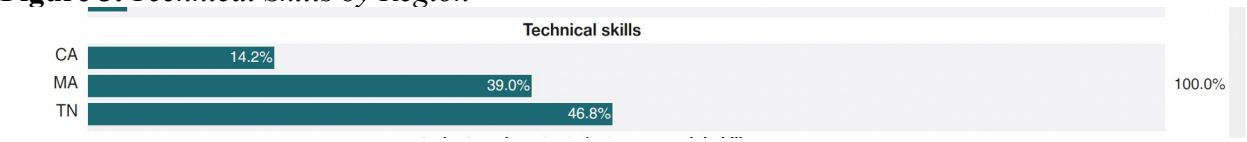
University/college participants accounted for 21.9%, while high school and school district participants each represented 20.8%. Elementary participants accounted for the smallest proportion, at 7.3% (see Figure 2). Thus, although technical skills were discussed across the PreK-14 continuum, they were considerably less prominent among participants associated with elementary grades.

Figure 2. *Technical Skills by Grade*

Normalized excerpt count for the selected code, grouped by descriptor.

Regional differences were even more pronounced. Tennessee accounted for 46.8% of technical-skills references, followed by Massachusetts at 39%, while California accounted for only 14.2% (see Figure 3). This indicates that technical skill development was substantially more prominent in how Tennessee and Massachusetts participants discussed IvE than in the California interviews.

Figure 3. *Technical Skills by Region*

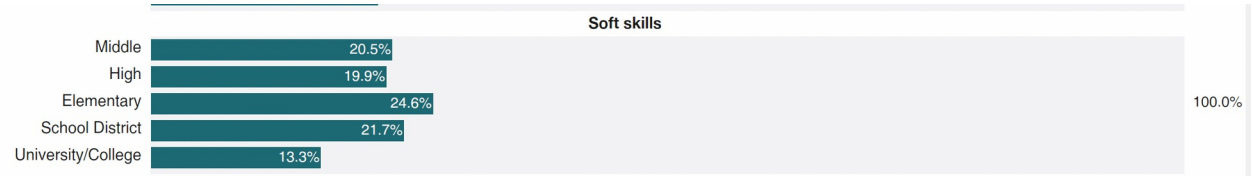


Normalized excerpt count for the selected code, grouped by descriptor.

When looking at soft skills as opposed to technical skills the distribution becomes more uniform across regions and grades. Almost even distribution across regions suggests that soft skill development is a near-universal goal for all of our partner organizations. Additionally, the near uniform distributions across grade level shows that these soft skills are important across the K-14 Pathway.

At the grade level, elementary participants accounted for the largest proportion of soft-skills references at 24.6%, followed by school district participants at 21.7%, middle school at 20.5%, and high school at 19.9%. University/college participants accounted for a somewhat smaller proportion at 13.3% (see Figure 4).

Figure 4. *Soft Skills by Grade*



Normalized excerpt count for the selected code, grouped by descriptor.

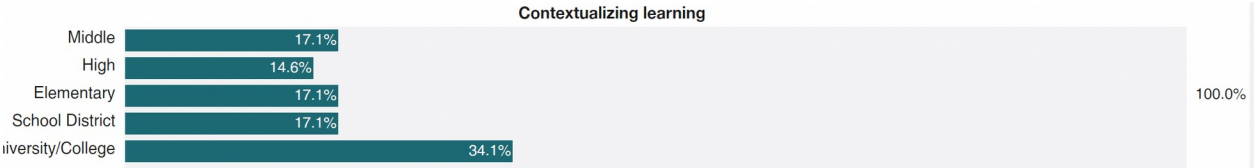
A similar pattern can be observed across regions. Massachusetts accounted for 37.2% of soft-skills references, California for 33.7%, and Tennessee for 29.1%. Although Massachusetts had the largest proportion, the relatively narrow range across regions contrasts with the much larger regional differences observed for technical skills (see Figure 5). In our study, the term "soft skills" was frequently mentioned by our participants and was used by the researchers to label a broader list of codes that include human, durable skills, including critical thinking, problem solving, building curiosity, creativity, among others.

Figure 5. *Soft Skills by Region*

Normalized excerpt count for the selected code, grouped by descriptor.

Participants also frequently conceptualized IvE as a way to contextualize learning by connecting academic content and skill development with authentic applications and real-world problems. Unlike soft skills, this theme showed clearer differences across grade levels. When viewed by grade level, it is clear that participants affiliated with universities mentioned contextualizing learning very often, more so than any other level, at 34.1% (see Figure 6).

Figure 6. *Contextualizing Learning by Grade*



Normalized excerpt count for the selected code, grouped by descriptor.

Additionally, participants from the California region mentioned contextualizing learning a bit more than the other two regions, with around 42.7% of the mentions, followed by Massachusetts at 33.9% and Tennessee at 23.5% (see Figure 7). Thus, although contextualizing learning was represented across all three regions, it was almost twice as prominent among California participants as among Tennessee participants.

Figure 7. *Contextualizing Learning by Region*

Normalized excerpt count for the selected code, grouped by descriptor.

Another important dimension of participants' conceptualizations of IvE involved changes in student mindset and identity. This category displayed one of the clearest differences across grade levels. Student mindset and identity shift was mentioned the most by the middle school level at 39.6%, followed closely by university/college at 36.3% (see Figure 8). The trend with university/college could be because adults are less likely to believe that they can become inventors. However, this trend is interesting to see that the middle school level is even higher.

Figure 8. *Student Mindset and Identity Shift by Grade Level*

Normalized excerpt count for the selected code, grouped by descriptor.

Another trend emerged when viewing the codes mentioning the adaptability of IvE. High schools had the most mentions with 35%, followed by university/college with 30.6% (see Figure 9). This is likely because high schools offer pathways such as AP, IB, and CTE. These pathways align with IvE curricula. For the university/college level, this makes sense because university curricula is typically more flexible than K-12, so professors are free to implement IvE into their classrooms.

Figure 9. *Adaptability of IvE by Grade Level*

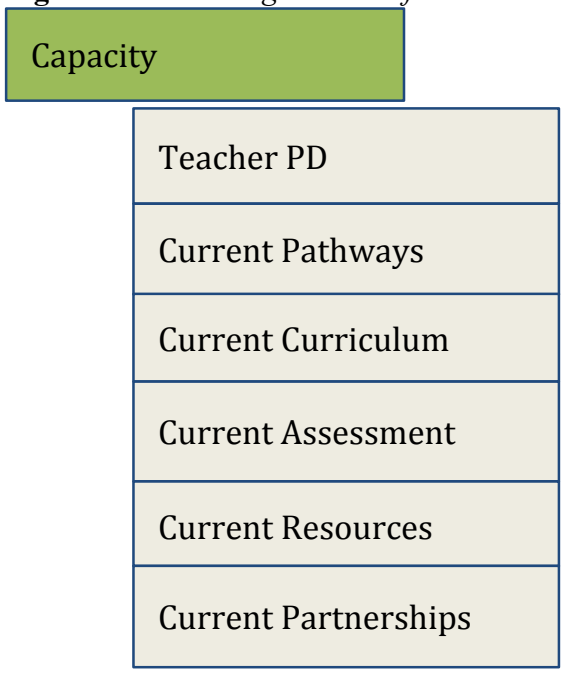
Normalized excerpt count for the selected code, grouped by descriptor.

Collectively, these findings suggest that regional partners conceptualized IvE as a multidimensional educational approach that combines durable and technical skill development with authentic, contextualized learning, shifts in students' identities as problem solvers and inventors, and flexibility across educational contexts. At the same time, differences in the distribution of particular codes demonstrate that the relative emphasis placed on these dimensions varies across regions and points along the PreK-14 pathway.

Overall Findings for Capacity

The domain of Capacity refers to current resources, supports, partnerships, and courses that promote IvE. We separated the category into six sub-categories: *Current Assessments, Teacher Professional Development, Current Curriculum, Current Partnerships, Current Pathways,* and *Current* Resources (see Figure 10).

Figure 10. *Code Organization for Future Goals*



Current Pathways. The Pathways category focuses on building and maintaining IvE

pathways for K-14. Each interviewee described their organization's pathway differently depending on their resources and focus. The Current Pathways domain included three subcategories: *Grade-Focused Pathways, CTE Pathways,* and *Building K-14 Pathways*.

Many organizations focused on implementing IvE at specific grade levels. Participant 5, an Industry Partner from California, commented:

> *"Most of our content that we create so far is really heavily geared towards 4th and 5th graders, really fourth graders. That's the … that's our bread and butter of school programs…"* [Participant 5, Industry Partner, CA]

Meanwhile, other organizations focused on older students. Participant 18, a university administrator from Massachusetts, said their organization is:

> *"... predominantly focused on undergraduate teaching and training…" [Participant 18, University Administrator, MA]*

Many participants reported working for a partner organization that focused on a specific subsection of the K-14 pathway. Furthermore, an outcome of IvE pathways is that sometimes specific groups of learners have an increase in participation. Participant 6 commented that implementing IvE increased enrollment among underrepresented learners:

> *"They also had a lot of problems with low enrollment of female students in these courses within the whole department. When I started, it was … roughly 7%. Any now we hover anywhere between 35 and 39% female…"* [Participant 6, High School Teacher, MA]

A common point of discussion in interviews was how pathways are built within school districts. Many participants described starting small, with either a specific class, grade level, or school for implementing IvE. Participant 19, a school district administrator from TN, commented:

> *"Here is started school-based at first, but then we actually expanded it out with the local university…"* [Participant 19, School District Administrator, TN]

There was a focus on piloting IvE in a particular location first before spreading it to other schools. Participant 14, a High School Principal in CA, similarly commented:

> *"I think we're going to start small there. And then maybe extend into all elementary schools…"* [Participant 14, High School Principal, CA]

Participants from the Tennessee school districts described building the school district's pathways from focusing on the high school to focusing on K-12. Participant 19 from Red Bank, TE, commented on how the school district's partnership with Volkswagen E-lab as a contributing factor, saying:

> *"... High School was one of six schools initially that got a … Volkswagen E-lab, VW E-lab. It's a Fab Lab. They were one of six original ones … I think that's about 2018, we were just focused on high schools. And as the years went by, we began to see that there was an, interest in middle and elementary. And then Red Bank Elementary added one…"* [Participant 19, School District Administrator, TN]]

Beyond IvE, Career and Technical Education (CTE) pathways play a key role in student education. Many interviewed educators commented on different types of career pathways in their high schools, with Participant 7 mentioning they had three different pathways, including a healthcare pathway. Meanwhile, Participant 14 shared:

> *"... an agricultural academy here at Cordova Academy, we have a culinary program, we have a media tech production program, we have a business academy, we have engineering. We have AVID, which stands for Advanced Placement Via Individual Determination … We also have robust visual and performing arts pathways."* [Participant 14, High School Principal, CA]

Teacher Professional Development. Training teachers to implement IvE is important but also challenging. In this section, we will discuss several examples of teacher professional development currently being implemented. The most common theme that emerged from the participants is that teachers learn best from hands-on experiences. For instance, Participant 14, a High School Principal in California, said:

> *" I think anytime that you, in a professional development, have your participants do what you're asking students or kids to do is the best way that you'll learn."* [Participant 14, High School Principal, CA]

More specifically, many participants referenced the invention sprints as ways to build excitement around IvE while also giving educators a look at what IvE really is and how to best support their students.

Another form of professional development is trainings for pre-service teachers. Several participants identified expanding class offerings centered on IvE for pre-service teachers as an opportunity to quickly ramp up the number of teachers trained in IvE. Participant 11 outlines this strategy from the perspective of universities saying:

> *"My colleague* [colleague's name] *has really been working on, the pre-service credentialing students at Stan State, and getting them to understand what invention education means, and how they can start bringing it into their classrooms when they get teaching jobs."* [Participant 11, University Professor, CA]

One thing that is universally necessary to implement IvE is strong support and backing from

administrators. This is also true with regard to professional development. Support from administration helps teachers fund and secure the specialized training needed to implement these classes. Participant 2, a university professor from CA, is especially thankful for the support from administration, saying they have a leader in their school who:

> *"…fully supports this, not just in spirit, but she helps by, you know, encouraging her teachers to attend our professional development center, or take field trips to our Fab Lab, and so that, you know, the teachers and students can."* [Participant 2, University Professor, CA]

Current Curriculum. The Current Curriculum subcategory describes how IvE is incorporated into existing curricula. Five topics emerged: *Types of Learning, State-Sponsored Curricula, Invention Education Competitions, Extracurricular Programs*, and *Class Offerings.*

Across several *Types of Learning* mentioned, participants most referenced hands-on, project-based, problem-based, inquiry-based, and design-centered approaches to IvE. Hands-on activities involved using tools and technology to design tangible products. Participant 22, an industry partner in California, described a marble-maze activity:

> *"One of the examples is our marble maze.… We laser cut cardboard and then they fold it and… use hot glue to create a marble maze."* [Participant 22, Industry partner, CA]

Project-based learning often involves collaborative work connected to students' interests or community needs. Participant 10, a Tennessee Middle School Principal, explained:

> *"Getting kids to work in teams, identify problems.… The kids created a mobile business. And some did that based on need, some did that based on personal interests, so they created in small groups*." [Participant 10, Middle School Principal, TN]

Problem-based learning similarly centered on presenting students with challenges and asking them to develop possible solutions. Participant 15, a California Elementary School teacher, stated:

> *"I might present a challenge to my students, and then they work through how they would solve that challenge.… By the time they get to the fifth year in the IB, it's supposed to be way more authentic, where… they might go out into the community."*[Participant 15, Elementary School Teacher, CA*]*

Participant 3, a Connecticut high school teacher, described an inquiry-based research lab for high school students:

> *"What we use is… AI technology to sift through molecular networks of molecules that are related to known antibiotics and try to identify other molecules that haven't been studied very much that could have antibiotic activity*."[Participant 3, High School Teacher, CT]

Participant 3 distinguished inquiry-based activities from "cookbook" labs. Inquiry-based labs involved less reliance on a predetermined process and greater student involvement in investigation and decision-making, although they also required more instructional time.

Design is a key aspect of many IvE learning methods, but design-centered instruction explicitly incorporates planning, iteration, and testing. Participant 13, a Massachusetts elementary teacher, described a Museum of Science curriculum that:

> *"Does follow the… engineering design process of… planning, and… coming up with some different iterations of a design, and testing it*." [Participant 13, Elementary School Teacher, MA]

Together, these excerpts demonstrate that IvE curricula use multiple forms of learning in which students identify problems, create products, conduct investigations, and test or revise their work. It is important to note that many of these *Types of Learning* overlap with each other in definition and practice, but a characteristic shared across all is an actively involved student.

Participants in California described *State-Sponsored Curricula* that aligned closely with IvE offerings. Participant 1, a California High School teacher, explained that their class was part of a Career Technical Education pathway:

> *"It's a CTE, like a Career Technical Education class, and there's different sectors. Our sector is manufacturing, and the pathway is product innovation and design."* [Participant 1, High School Teacher, CA*]*

The CTE pathway provided an established curricular structure through which product design and invention-related activities could be incorporated into formal course offerings. CTE and IB systems were often mentioned by California Participants, leaving us with the conclusion that state-sponsored curricula in this region enhanced capacity for supporting IvE, rather than limiting it, like participants experienced in other regions.

In a more fluid way, *IvE competitions* were implemented as a way to organize design challenges across classrooms and schools. These competitions allowed flexibility for school and grade level and targeted students' competitive nature to encourage IvE. Participant 9, a Tennessee industry partner, stated:

> *"Our teachers release design challenges all the time. I just do one district-wide one where all the students are kind of… competing against each other. So there are tons of different design challenges going on."*[Participant 9, Industry Partner, TN]

This district-wide structure allowed students from different classrooms and schools to participate in a shared design activity while building upon challenges already conducted by individual teachers. Also noted by this participant was the importance of making competitions accessible across the K-12 spectrum. Competitions tend to focus more on the higher grade levels, and creating competition opportunities for elementary students requires more extensive planning to cater to their interests and abilities.

IvE *Extracurricular Programs*, including after-school programs, summer camps, and clubs, appeared in 14 excerpts. Participant 22, an industry partner from California, described a week-long makerspace summer camp:

> *"We have… summer camps.… That's a week that students come, fifth to eighth graders, to our makerspace, and they have a project, and they learn CAD, they learn how to design on the computer, and then 3D print, and laser cut, and solder."*[Participant 22, Industry Partner, CA]

Participant 16, a CTE director in Massachusetts, described an extracurricular club focused on sustainable building design:

> *"We also have started a high performance building innovators club.… Students meet outside of the school day, either in the morning or the afternoon, and in the summer they actually can earn a stipend based on their participation.… The students are designing and developing what we hope is a net zero ADU*."
> [Participant 16, CTE Director, MA]

These examples demonstrate how summer programs and clubs can extend IvE beyond the regular school day. These programs can also provide students with more time to use specialized tools, develop technical skills, and complete longer-term projects. Even when in-school capacity is limited, *Extracurricular Programs* can fill the gap in many forms.

The extent to which IvE was incorporated into formal class offerings varied across schools. Participant 23, a Tennessee high school teacher, described IvE as primarily dependent on the initiative of individual teachers:

> *"Invention Education right now in my area happens on… a micro level within classrooms where teachers are motivated to do so and to plan a project centered around invention."*[Participant 23, High School Teacher, TN]

In other schools, IvE was incorporated more consistently into engineering courses. Participant 6,

a Massachusetts high school teacher, explained:

> *"I have engineering classes, and within the engineering classes… let's say I have five units per semester. Typically, three to four of those units are Invention Ed-based."* [Participant 6, High School Teacher, MA]

A resounding theme in the *Current Curricula* subcategory, as well as the *Capacity* grouping as a whole, was that IvE implementation is highly context-dependent. What works for some schools may not work for others.

Yet another method, the short exploratory rotations described by Participant 20, a Massachusetts high school administrator, allowed students to experience several subjects before selecting areas of greater interest:

> *"The first rotations are three days long. So we call that speed dating.… Second quarter, you get to choose five rotations that you thought were really cool, that you were interested in, and out of those five, we try to accommodate three of them for you to spend two weeks in*." [Participant 20, High School Administrator, MA]

Participants also noted that IvE-related subjects compete with other curricular priorities. Participant 4, a Science director from Massachusetts, stated:

> *"We prioritize literacy, numeracy, and a lot of resources and energy go into supporting those goals, which makes a lot of sense to me. But sometimes that means… science or social studies, especially in the younger grades, is less of a priority."* [Participant 4, Science Director, MA]

The most consistent takeaway across the *Capacity* section is a need for context in conversation about IvE implementation. The schedules of different schools have different opportunities for IvE. In the excerpts above, short rotations, a standalone class, micro-projects, and extracurricular activities were all mentioned, but those were far from all that were captured by the participants. Some schools had their own lab and multiple lab-specific teachers, where others had teacher shortages and used labs from a neighboring university. Across these accounts, IvE took on a range of expressions, styles, and interpretations. Its place in the curriculum depended on course structures, teacher involvement, scheduling, available programs, and competing instructional priorities.

Current Partnerships. The Current Partnerships subcategory details partnerships that help facilitate IvE within a region. These partnerships operate at the school, district, and regional levels. Two topics emerged within this subcategory: *Industry/Community Partnerships* and

Academic Partnerships. This distinction highlights differences in the partnerships available across the target regions.

Industry/Community Partnerships often reflect the cultural and economic environment of a region. In the Tennessee region, participants frequently discussed a partnership with Volkswagen (VW). VW ELabs, also known as Fab Labs, have been implemented in nearly every school in one district and are spreading into others. Participant 25, a high school administrator from Tennessee, explained:

> *"Our elementary and the middle school that feed to… to our high school… we all have E-Labs, digital fabrication labs*." [Participant 25, High School Administrator, TN]

Manufacturing, distribution, and supply-chain industries are among the region's major industries, providing context for the VW partnership. Participant 9, a Tennessee industry partner, described how the VW partnership began:

> *"Volkswagen made contact with our local district, Hamilton County, and said that they had money that they would love to use to improve our district.… They have poured money into our labs. So they fund every lab*." [Participant 9, Industry Partner, TN]

The VW partnership demonstrates how a local industry partner can provide direct financial and material support for IvE implementation across a district. Participants in other regions also described partnerships connected to local industries. Participant 16, a CTE director in Massachusetts, reported:

> *"We have partnerships with, in the construction industry, we have partnerships with trade unions, carpentry, trade union, electrical, as well as multiple large-scale construction organizations that build large-scale projects."[Participant 16]*

In the California region, museum partnerships were particularly prominent. Participants 7 and 14 identified partnerships with museums. Participant 7, an administrator for a virtual high school in California, described seeking museum partnerships that could provide makerspace opportunities:

> *"[museum name]is widely underutilized just because… it's a museum and they want kids to come in."*[Participant 7, High School Administrator, CA]

California was also the only region in which the interviews included a mix of virtual and in-person schools. Participants described museums as physical locations where students, including those enrolled virtually, could access hands-on IvE activities and resources.

The next topic in the Current Partnerships subcategory was *Academic Partnerships*. *Academic partnerships* primarily focused on partnerships with local universities, including

access to college courses, equipment usage, and research projects.

Local university partnerships could provide facilities, equipment, and project opportunities that schools could not independently offer. Participant 7 explained:

> *"[The local college] has lab space that… if we have kids come to the site with us, then they're able to kind of have all of that stuff at their fingertips, rather than us trying to invest in creating or buying all those materials."*[Participant 7, High School Administrator, CA]

Local university partnerships, like the one mentioned by Participant 7, are especially useful in virtual schools, small schools, or schools lacking in staff support. A local university can act as an extension of a school's space, staff, and resources to support IvE offerings. Dual enrollment programs specifically can provide a structure to these types of Academic Partnerships:

> *"We also partner with the community colleges on their CTE certificate program. So we help support students in like a Dual Enrollment learning environment where [local college] works with us to provide sections for students so that they can meet that certificate requirement*." [Participant 7, High School Administrator, CA]

Dual Enrollment, which often allows high schoolers to obtain high school and college credit simultaneously, provides a unique avenue for meeting school standards and exposing students to a wider range of classes. Already, some Dual Enrollment classes have started to incorporate IvE, but it remains an opportunity for further IvE Capacity building.

Aside from explicit Dual Enrollment courses, local universities can give students access to projects and research outside typical high school offerings. A Connecticut High School Teacher, Participant 3, described connecting university researchers with high school students:

> *"I act as a conduit between the professors that have a project or need some grunt work done at [local university], and the students at the high school who want something to work on."* [Participant 3, High School Teacher, CT]

This arrangement connected university projects with high school students seeking project experience while allowing schools to draw upon resources and expertise available through local higher education institutions. It is important to note that 7 out of the 23 excerpts captured surrounding partnerships with local universities were from Participant 3. This Participant, who described themselves in the excerpt above as "a conduit," provides an interesting insight into the role of teachers in their capacity to form *Academic Partnerships.*

Current Assessments. The assessment category contains examples of current assessment

approaches and tools used by teachers. Teachers use a variety of assessments to evaluate student learning in IvE, which we grouped under the subcategories of *Reflective & Formative Assessment (e.g., student surveys, reflections, peer feedback, Exit Ticket*), Traditional Assessment (e.g., exams, quizzes), *End-of-Course Showcase*, and *Real World Impact Assessment (i.e., evidence of student learning outside class).*

One of the most frequently mentioned ways of assessing students is with the help of custom rubrics. Participant 8, an Elementary School reading Specialist, shared:

> *"There's a rubric that most grade levels have created to assess those different skills that they're looking to assess for that unit."* [Participant 8, Elementary School reading Specialist]

Another way teachers assess their students is by looking at their portfolio of work. Assessing an entire portfolio can be aided by tools such as FabFolio. FabFolio is an online, AI-powered educational platform designed to measure student learning in certain skills. To do this, students build their portfolio and earn micro-credentials verifying their mastery of soft and technical skills. Participant 19 shared:

> *"…students submit their projects online into this, lack of a better term, online portfolio. And then students go in, and they have to submit evidence, to show that they've hit that criteria of whatever technical or technical skills are being evaluated, and whatever durable or essential skills are being evaluated*" [Participant 19, School District Administrator, TN]

Several participants choose to have their students present their inventions to the community at the end of the class. This serves several purposes. First, it gives teachers a physical deadline which students must have their work in by. This allows the teachers to use previously mentioned strategies such as rubrics to assess the students work and assign grades. The exposition also brings in support for the program. Oftentimes, school administration and key stakeholders will see the work students are doing in these classes and increase support for the programs. As Participant 6, a high school teacher from MA, shared:

> *"…inviting administration into… product showcases, and… having them participate in judging student work. And they got to see firsthand the increased activity and interest that the students had. Prior to when we had put these things in place"* [Participant 6, High School Teacher, MA].

Current Resources. The Resources subcategory encompasses the different types of resources regional partners have access to. Each regional partner had differing types of resources

to help promote IvE, so many factors were considered for this subcategory, ranging from *Staffing* and *Material Resources* to *Financial Allocations* and *Logistical Constraints*.

One prominent theme was how teachers currently allocate budget to funding their classrooms. Many teachers mentioned the use of outside grants to fund their access to certain resources. Participant 1, a high school teacher from TN, said:

> *"I have a grant. So, we have, like, 80 … it's a grant that, gives us $900 per student per year, so we're really lucky to have, like, a lot of resources. So, like, I can really buy whatever I want. The limitation is that I can't buy a new room … But I have, in a closet that I removed all the cabinetry, a 4x4 foot, table CNC machine mill. And then … I've stacked printers onto carts so that they are vertical … So, I think, like, if you have money, there's really cool tools"* [Participant 1, High School Teacher, TN]

In addition, flexibility to use the funding as teachers want was seen as an added benefit. Participant 23, mentions the use of flexible funding from Volkswagen, saying:

> *"Something that's been really amazing from Volkswagen is that they have given, really, that flexibility to … utilize those funds in the way that's gonna make the most impact"* [Participant 23, High School Teacher, TN]

Many educators detailed classroom schedules and programming timelines during the interviews. Classroom time engaging with IvE varied widely. Participant 24, a university professor from CA, says their high school classroom will likely have an upcoming fall schedule with a *"... hybrid Monday, Wednesday, Friday with 3 hybrid and like online."* Meanwhile, other classes meet on a less regular basis. Participant 24, a university professor from CA, commented:

> *"I'll be doing engineering with all the students in the school, seeing each class at least once a week"* [Participant 24, University Professor, CA]

In addition to school schedules, many educators also mentioned the time taken to build IvE pathways. Often, school pathway building was on the scale of years. Participant 3 says it is a:

> *"…15-year process at least" and continued on to detail that "the whole thing [IvE] started in 2008. It was really … getting into gear in 2011. We started making visits to the different companies in, I'd say, 2013 or so" [Participant 3, High School Teacher, CT]*

Time to build pathways was seen as a resource, as a school needed to invest several years into building an IvE model.

Current material and physical resources to promote IvE were another common focus of Capacity. One form of this resource was the use of Fab labs and eLabs within schools.

Participant 11 describes the Fab Lab as:

> *"... a type of makerspace that typically operates with digital technology. So a lot of the common equipment is 3D printers, laser cutters. You might have like CNC mills or CNC routing capabilities." Another form of resource is Easy-use kits for teachers with little background in IvE. These kits were described by Participant 2 as "each kit has a different theme. But they're standalone curriculum with all the materials that are literally in a kit, a giant box" [Participant 11, University Professor, CA]*

Outside of labs and kits, providing students with physical spaces to meet to learn and participate in IvE became another interview focus. Participant 10 describes use of part of their eLab as:

> *"We call it the TED talk space. So kids start in there, they hear the presentation about the problem and they start ideating in that room and then they work through the eLab spaces"* [Participant 10, Elementary School Leader, TN]

Meanwhile, Participant 19, a high school leader from TN, says their organization has *"... Future Ready Centers, where students are able to come half-day from their local high schools, to participate in robust CTE."* Providing physical locations for students to learn acts as another resource to enable student learning.

Another emerging theme in several interviews was the role that school administration, including principals and superintendents, had on teacher access to resources. The more well-versed school administration was with IvE curriculum, the more support teachers had to implement IvE. Participant 15 shared:

> *"Our superintendent is very, supportive as well. So, she … was familiar with invention education from a prior district. And so, when I reached out to her, to say that I wanted to bring invention education into my classroom, and then I wanted to be able to take kids to the California Invention Convention. She was very supportive with money and resources" [Participant 15, Elementary School Teacher, CA]*

Superintendents played a key role in several teachers' ability to access resources and support to implement IvE. Similarly, teachers comfortable with IvE, were commented as better able to tailor the current curriculum to their students. Participant 17 said:

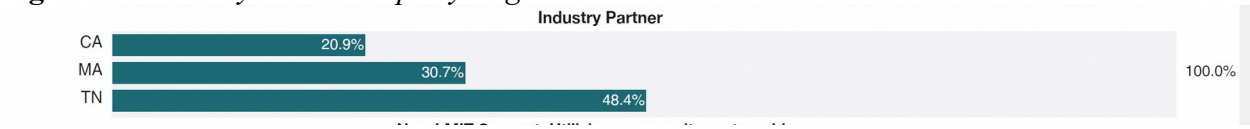
> *"So, it really became theirs, as opposed to, let's just follow what the curriculum says. So, the more a teacher knows about the curriculum, the better they can adjust for their … their kids" [Participant 17, Elementary School Leader, MA]*

Key Trends for Capacity by Region, Role, and Grade Level

Key Trends by Region. Several trends were seen within *Industry and Academic Partnerships* among regional partners. While there was a relatively even distribution of

partnership mentions among the different regions, a more skewed distribution emerges when analyzing *Industry Partnerships* and *Academic Partnerships* (see Figures 10,11).

Figure 10. *Industry Partnerships by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

Figure 11. *Academic Partnerships by Region*

Normalized excerpt count for the selected code, grouped by descriptor.

The Tennessee region had the highest normalized percentage of participant responses for Industry-specific partnerships at 48.4%, likely due to the prominent Volkswagen (VW) partnership in the region, which sponsors ELabs in many of the schools. Specifically,10 out of the 20 Industry Partnership excerpts from the Tennessee region mentioned VW, and the remaining 10 were various unique partnerships. However, in the Massachusetts and California regions, no dominant partnerships emerged, and the same partnership was never mentioned by two participants more than twice.

When looking at the graph of *Academic Partnership* normalized by region, California and Massachusetts take the majority, with 44.1% and 44.4% of the excerpts, respectively. Tennessee *Academic Partnership*, on the other hand, only included 11.4% of the mentions.

This data points to an important conclusion: different regions have different strengths regarding partnership formation. Tennessee made use of the thriving local manufacturing industry to facilitate IvE; California and Massachusetts took advantage of the abundance of local universities. As these regions progress, it will be important to be cognizant of the different challenges that come with scalability in regions that depend more on certain types of partnerships. An *Industry Partnership* may only be as beneficial as local resources allow before a bottleneck forms, and likewise for university availability and funding.

On a similar note, it may be worth investigating if academic and industry partnerships compete with each other in any way. Although, more likely than competing, they seem to fulfil

similar needs within a school. In Tennessee, for example, the existence of VW ELabs decreases the need to rely on local universities for makerspace and material access.

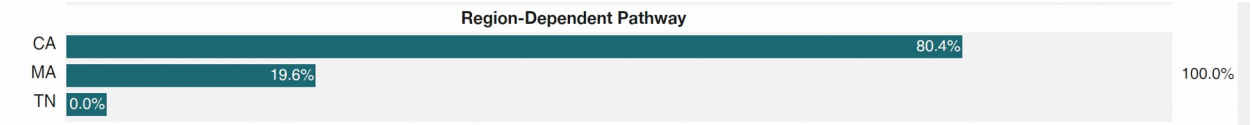
Another point of current capacity discussion during interviews was CTE Pathways. The Tennessee region had over 50% of the mentions of schools offering CTE pathways. Both California and Massachusetts had under 25% (see Figure 12).

Figure 12. *CTE Pathways by Region*

Normalized excerpt count for the selected code, grouped by descriptor.

On the other hand, the category of *Region-Dependent Pathway* was skewed towards California, with more than 80% of references, which could suggest that California has a curriculum that is unique to the region, where Massachusetts and Tennessee are more standardized. Furthermore, within the *Current Curriculum* subcategory, the only references to state-sponsored curriculum are 100% from the state of California for their PORTS program and California CTE/IB requirements, which helped support IvE initiatives (see Figure 13).

Figure 13. *Region-Dependent Pathways by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

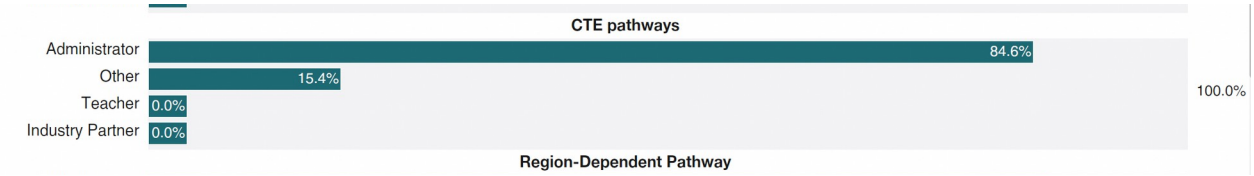
Another differing theme among regions was forms of assessment discussed. Among California participants, traditional forms of assessment were mentioned at 62.1% compared to 37.9% among Massachusetts participants and no mentions among Tennessee participants. Meanwhile, Tennessee participants had the highest reference numbers for reflective and formative assessments at 39.8% compared to the other two states. End-of-course showcases and real-world impact assessments were exclusively mentioned by participants from Massachusetts (see Figure 14). This shows potential differences in regional focuses. While California had a higher discussion of traditional assessments, it had a relatively lower mention of reflective and formative assessments, which was the exact opposite of Tennessee.

Figure 14. *Traditional Forms of Assessment by Region*

Normalized excerpt count for the selected code, grouped by descriptor.

Key Trends by Role. CTE pathways are a point of interest to administrators. There was a significant spike of 84.6% in the normalized percentage of participant responses among administrators for referencing CTE pathways. Meanwhile, there were relatively more evenly distributed references to K-12 pathways and building pathways between administrators and teachers. This may show a difference in priorities and focus between administrators and teachers (see Figure 15).

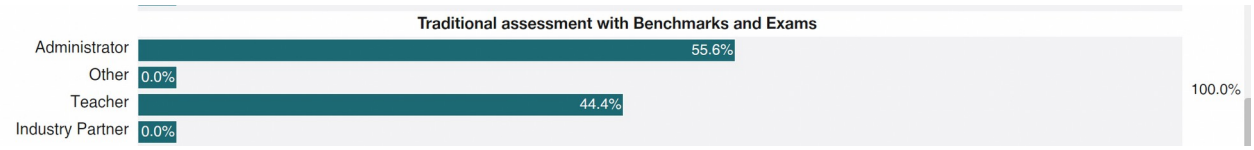
Figure 15. *CTE Pathways by Role*



Normalized excerpt count for the selected code, grouped by descriptor.

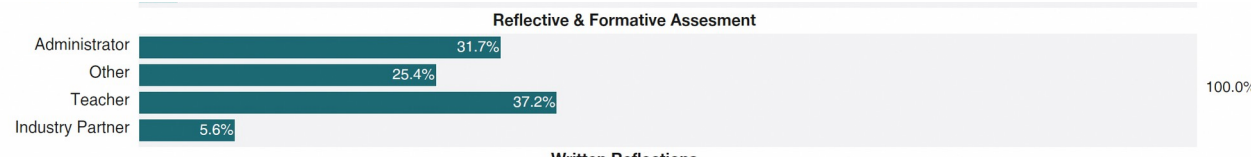
Administrators and teachers had a spike in mentions of traditional assessment. Administrators mentioned traditional forms of assessment 55.6% of the time and teachers mentioned this form of assessment 44.4%. This is in comparison to other forms of assessment, such as reflective and formative assessment, which was mentioned by administrators, teachers, and others in similar distributions. Meanwhile, only teachers mentioned other forms of assessment, such as the end-of-showcase and the real-world impact of assessments. This may imply that administrators and teachers have different assessment priorities (see Figures 16,17) or focus on different outcomes, potentially due to state accountability mechanisms. While interviews with administrators focused on traditional forms of assessment, interviews with teachers had multiple assessment types discussed.

Figure 16. *Traditional Forms of Assessment by Role*



Normalized excerpt count for the selected code, grouped by descriptor.

Figure 17. *Reflective and Formative Assessment by Role*



Normalized excerpt count for the selected code, grouped by descriptor.

Discussion of professional development also differed based on role. Current professional development capacity was mainly mentioned by Teachers at 43.4% and the Other category at 27.1%. Teachers also had the highest reference rate of 62.5% for training methods and resources. Given that professional development typically focuses on teachers, it would be expected for the issue to be referenced more among this population of participants (see Figure 18).

Figure 18. *Training Methods and Resources by Role*

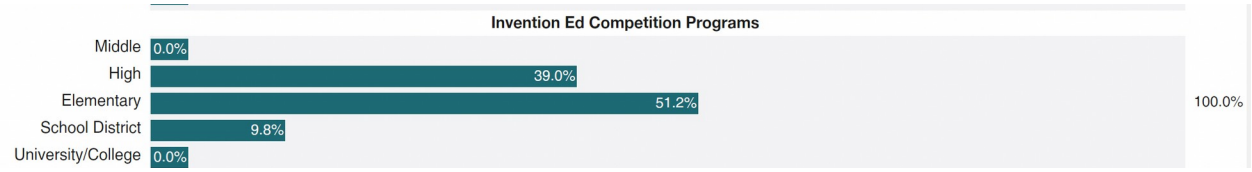
Normalized excerpt count for the selected code, grouped by descriptor.

Key Trends by Grade Level. One trend that emerged from grade level analysis was differences in types of learning and learning focus areas. Elementary school teachers from California were the only participants to mention problem-based learning as a type of learning focus in their curriculum. Meanwhile, lab-based enquiry was exclusively mentioned by high school participants, with a majority of the participants identifying as teachers. Project-based learning was mentioned across all regions, with respondents primarily from middle (66.7%) and high (23.8%) schools. The frequency of mentions of hands-on learning was equal for high schools and school districts with 38.7%, followed by universities/colleges with 22.6%.

This may indicate different learning focus areas across grade levels. Skills like lab/inquiry-based and project-based learning skewed more towards middle and high school respondents, while problem-based learning skewed more towards elementary school respondents.

IvE competition programs were mentioned by all state participants within 7% of mentions of each other. The majority of competitions were mentioned by High School (39.0%) and Elementary School (51.2%) participants (see Figure 19). This may suggest a gap in competition offerings among Middle School and University/College if all states mentioned competitions, but only certain partner roles mentioned competitions.

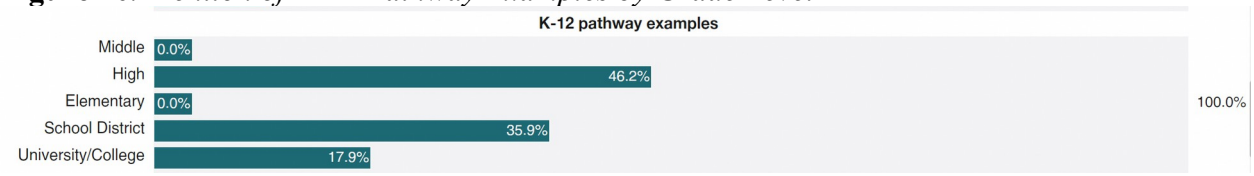
Figure 19. *Invention Education Competition Program Mentions by Grade Level*



Normalized excerpt count for the selected code, grouped by descriptor.

High school was the most frequently referenced grade level for K-12 pathway examples (46.2%), followed by the School District level (35.9%). CTE pathway building was only mentioned by the School District (66.7%) and High School participants (33.3%), reflecting participants' emphasis on preparing students for college and careers (see Figures 20, 21).

Figure 20. *Mention of K-12 Pathway Examples by Grade Level*



Normalized excerpt count for the selected code, grouped by descriptor.

Figure 21. *Mention of CTE Pathways by Grade Level*

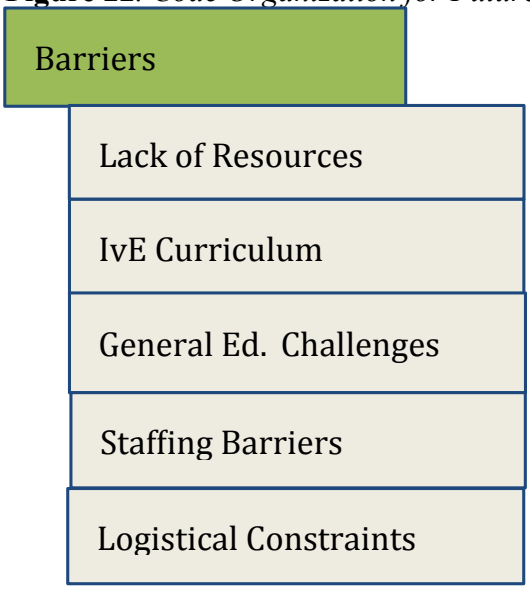
Normalized excerpt count for the selected code, grouped by descriptor.

Another interesting trend was observed for elementary educators, where they discussed integrating IvE with humanities and social sciences (100%), suggesting that invention education begins through interdisciplinary learning before expanding into structured pathways in later grades.

Overall Findings for Barriers

Participants identified a range of interconnected barriers that may limit the implementation and sustainability of Invention Education (IvE). We identified several categories of Barriers, including *Lack of Resources, IvE Curriculum, General Education Challenges, Staffing Barriers,* and *School Logistical Constraints* (see Figure 22). Although these challenges varied by region, participant role, and educational level, the findings illustrate how barriers operating at the classroom, school, and system levels can collectively shape access to and implementation of IvE.

Figure 22. *Code Organization for Future Goals*



Lack of Resources. The *Lack of Resources* subcategory describes limitations in the external, financial, and material support available for IvE implementation. Three topics emerged: *Community Support, Budget/Funding,* and *Physical Resources. Community Support* and *Physical Resources* appeared in six and seven excerpts, respectively, while *Budget/Funding* appeared most frequently, with 21 excerpts. The overlap among these topics suggests that funding limitations can restrict access to physical resources, while low community engagement may reduce the visibility and support needed to obtain additional funding.

The *Community Support* barrier refers to limited community participation in IvE programs and low public visibility of student work. Participant 1, a California high school teacher, described challenges with parent engagement and recognition within the wider school district:

> *"I think that one of the struggles my school is currently looking at is… community engagement… And it's really low… parent engagement. When the kids are doing these big things, especially at a school where we don't get as much… recognition as some of the other schools in our district because of our scores… like math and English scores are not great all the time in a school where 30% of the students are English language learners."* [Participant 1, High School Teacher, CA]

This excerpt connects limited community engagement with broader issues of school visibility. Even when students complete substantial projects, low parent participation and the school's academic reputation may make it more difficult for those accomplishments to receive recognition.

Budget/Funding was the most frequently identified topic within *Lack of Resources*, appearing in 21 excerpts. Participants described difficulties purchasing consumable materials, replacing technology, compensating faculty, and offering additional courses or extracurricular programs. Participant 23, a Tennessee high school teacher, described being unable to replace equipment or purchase the materials necessary for projects:

> *"I don't have the money to replace it, and I don't have the consumable… materials that are needed in order to… tackle a project like this."*[Participant 23, High School Teacher, TN]

Five additional excerpts identified consumable materials or technology as budget-related barriers, indicating that funding affects both the initial purchase and continued maintenance of IvE resources.

Participants also discussed the cost of compensating faculty for professional development, curriculum design, and work performed outside their regular responsibilities. Participant 11, a

professor in California, explained that faculty members were completing IvE-related work without the reassigned time normally provided for institutional projects:

> *"One of the things at [college] that typically happens for faculty is, like, we get buyout time or reassign time for projects that we're doing. And that costs money. So if we need more faculty… a lot of it right now is just service… we're doing it for free."* [Participant 11, Professor, CA]

Funding limitations also affected schools' ability to offer additional courses. Participant 3, a Connecticut high school teacher, described contractual requirements for compensating teachers who take on extra classes:

> *"In our contract setup, if a teacher teaches an extra course, they get a certain percentage of their salary extra, and because of that contractual constraint, we would be… unable to teach extra courses, even if we wanted to, without compensation."[Participant 3, High School Teacher, CT]*

Participant 21, school district administrator in California, described a school attempting to rebuild science, elective, and extracurricular offerings while facing ongoing fiscal limitations:

> *"And they're… in a position where there are no after-school activities. Science is not offered daily across all the different grade levels, there are no electives, there's no after-school program, so there are a lot of things that we're gonna have to, like, rebuild, you know, all while in the middle of a fiscal situation*."[Participant 21, School District Administrator, CA]

Together, these accounts show that budget constraints affect more than materials. They also limit faculty participation, course availability, professional development, and students' access to IvE outside the regular school day.

Physical Resources includes access to materials, facilities, equipment, and transportation. Excerpts in this topic were frequently coded alongside *Budget/Funding,* indicating that material barriers were often connected to financial limitations. Even efforts to reduce costs by using recycled materials still required substantial resources. Participant 15 explained their experience as a California elementary school teacher:

> *"Money, there's all, you know, even when we tell students, like, 'Oh, just use recycled materials to make these inventions,' like, it still takes a lot of… it takes a lot of resources."* [Participant 15, Elementary School Teacher, CA]

Transportation was another physical resource barrier, although it appeared specifically in an interview involving a virtual school in California. Because the school served students across a large geographic area without a traditional shared campus or school transportation system, access to physical IvE spaces became both a facility and transportation issue. Participant 7, an

administrator for a California virtual high school, stated:

> *"So it's really hard to invest in space when, again, you're covering a nine-county spread. So then it becomes an access issue. So how do you help students who are in Contra Costa get all the way over to, you know, Folsom, which is a long drive?"* [Participant 7, High School Principal, CA]

Schools with physical campuses did not identify transportation as a significant IvE barrier in the available excerpts. This suggests that the relevance of transportation may depend on school structure and the geographic distribution of students.

Though many participants mentioned budget and funding limitations, or other resource barriers that linked back to funding, the issue was often not the money itself, but the freedom it offered. With additional funding, participants would be able to overcome contractual constraints and invest more in professional development. As an aside, it is important to note that though fiscal investment in material resources and space were mentioned by several participants, others shared that finances weren't a barrier at all, and that they could implement IvE on a very limited budget. It is worth investigating the financial means of the organizations of participants that stated funding as a significant barrier versus those who didn't mention it.

Either way, the most important takeaway from the *Lack of Resources* subcategory is that teachers should be paid for their work. IvE growing to be associated with teachers working against contractual constraints and without compensation is not a practice that should become standardized or tolerated. Setting this kind of standard early on in the programs can set a program back years. It is best to wait until there is enough funding to properly pay teachers for training before implementing IvE, so that when the program grows, it will thrive.

IvE Curriculum. The *IvE Curriculum* subcategory describes challenges associated with integrating IvE into school curricula. The primary topics represented in these excerpts were *Problem Identification* and *Assessment*.

Participants described *Problem Identification* as an early and particularly difficult stage of the invention process. IvE often encourages students to identify cultural, community, or personal problems that matter to them. However, students may struggle to generate ideas or recognize problems that could become the basis of an invention project.

Participant 1, a California high school teacher, described problem identification as the most difficult part of the process, despite having the resources to support students after a problem

was selected:

> *"If every kid could come in with, like, just lots of ideas… that's all I need. Like, that's the hardest part… Coming up with the problem is the hardest part. I can do all the things that, like, once we have the problem, I can help find the mentors, I can help find the subject matter experts, we can do research, we can do drawings… that all… we're already really good at… Kids have a hard time being creative."[*Participant 1, High School Teacher, CA]

Participant 25, a Tennessee high school administrator, similarly noted that students had fewer opportunities to identify problems earlier in high school:

> *"The opportunity to identify problems. I wouldn't say there's a lot of that… open opportunity.… Seniors, yes. But where can that fit, you know, earlier kind of in our curriculum or opportunities?"* [Participant 25, High School Administrator, TN]

These excerpts suggest that students may benefit from earlier and more frequent exposure to problem identification. Greater familiarity with the invention process may help students become more comfortable recognizing problems and developing their own project ideas.

Participants also identified a lack of IvE-specific *Assessment* methods. Teachers described uncertainty about how to evaluate invention-related skills and a continued reliance on traditional measures of memory and factual recall. Participant 2, a professor from California, explained:

> *"Teachers always struggle with what assessment could or should look like. And so… it's still very popular just to… assess memory and recall facts. Like, that's the easiest thing to assess."[*Participant 2, Professor, CA]

Confusion about assessment appeared in eight excerpts. Common concerns included reliance on traditional measures, uncertainty about expected deliverables, and difficulty evaluating skills that may not produce a single correct answer. Participant 23, a Tennessee high school teacher, shared:

> *"How are we going to assess essential skills?"* [Participant 23, High School Teacher, TN]

Participant 10, an elementary school administrator from Tennessee, contrasted traditional standards-based assessment with questions about how to evaluate students' problem-solving potential:

> *"How many standards can you cram into a kid's head in one day?"*[Participant 10, Elementary School Administrator, TN]

Participant 10 then described a shift toward a different question:

> *"We continue to underestimate kids' potential to solve problems or do hard things… but now… how do you level it up?"* [Participant 10, Elementary School Administrator, TN]

These excerpts reflect a broader assessment challenge within IvE; Teachers may recognize students' creativity, collaboration, and problem-solving abilities but lack consistent measures for evaluating those skills or demonstrating progression over time. If IvE-specific assessment measures are not developed and spread, the progress of IvE in general will be incredibly difficult to measure, and it will lose traction in the long run. Changing the existing educational infrastructure requires friction and a significant activation energy from teachers and staff, making the *Assessment* barrier of high importance for convincing teachers IvE is worth it.

General Educational Challenges. The *General Educational Challenges* subcategory describes barriers arising from broader conditions in education rather than from the capacity of an individual IvE program. The topics represented in these excerpts include *AI and Technology Uncertainty* and *Competing Academic Priorities.*

Participants expressed uncertainty about the effects of AI on students' future careers, academic interests, and durable skills. Participant 6, a high school teacher from Massachusetts, described concerns about changes in student enrollment in computer science:

> *"Everybody right now is freaking out about AI. Especially the computer science people. Some districts are seeing a shift in computer science numbers in terms of signups*."[Participant 6, high School Teacher, MA]

Other participants focused on the possible effects of AI on creativity. Participant 13, a Massachusetts elementary teacher, stated:

> *"I see that creative brain… shutting down.… In this age of AI… I worry that that will happen more and more*."[Participant 13, Elementary School Teacher, MA]

Concerns about AI overlapped with broader concerns about social media and technology. Participant 20, a school district administrator in Massachusetts, described the influence of online validation and social comparison:

> *"Social media as a whole is a huge disparate.… They feel the need to get likes.… It's just this whole make-believe cyber world they've created*."[Participant 20, School District Administrator, MA]

The data did not directly identify phone or technology use as a barrier to classroom instruction.

Instead, participants focused on uncertainty about how AI and social media may affect students' creativity, career decisions, self-perception, and understanding of the world. These concerns present a broader educational challenge and may increase the perceived importance of IvE activities that emphasize creativity, problem-solving, and engagement with physical or community-based problems.

Participants also described competition between IvE or Career Technical Education offerings and courses perceived as more valuable for college preparation. Participant 16, a school district director from Massachusetts, explained that some students prioritize Advanced Placement courses:

> *"There's a large contingent of students… that have a belief that taking as many AP classes as possible is their chosen route*."[Participant 16, School District Administrator, MA]

This preference may reduce student participation in IvE or CTE courses, particularly when students and families view Advanced Placement enrollment as more closely connected to college admissions. In this context, IvE programs have to compete not only for school resources and scheduling space but also for recognition as academically valuable course options.

Staffing Barriers. Staffing Barriers captures challenges related to the people required to implement and sustain IvE programs. Rather than focusing primarily on physical or financial resources, this category encompasses barriers associated with teachers, administrators, mentors, facilitators, and other personnel whose knowledge, capacity, confidence, and support influence IvE implementation. Participants described difficulties gaining teacher and administrative buy-in, securing appropriately trained staff and subject-matter experts, coordinating professional development, and ensuring that educators have the time, skills, and willingness needed to facilitate invention-based learning. Within Staffing Barriers, we identified four subcategories: *Challenges Obtaining/Convincing Staff, Training-Related Challenges, Teacher Capacities, and School Logistical Constraints.*

The subcategory of *Challenges Obtaining/Convincing Staff* encompasses difficulties securing the personnel, expertise, and organizational support outside of school needed to implement IvE. Participants discussed the challenge of helping teachers and administrators understand the educational value of IvE, gaining administrative approval, and identifying mentors or subject-matter experts who can support students' projects.

The most recurring theme was that school leaders and educators may initially misunderstand IvE. A few participants noted that, for school leaders, it was by focusing on its visible products rather than its broader educational purpose. Participant 9 explained:

> *"I think one of our biggest challenges sometimes can be to help a principal who doesn't know about our model understand that it's important. Because it is very easy for them to look at the cool equipment and go, 'Oh, kids make t-shirts*." [Participant 9, School District Administrator, TN]

This misunderstanding was not limited to administrators. Several participants also described the need to expose teachers to IvE before they could fully appreciate how it differs from traditional instruction. For instance, Participant 2, a university Professor in California, described creating opportunities for teachers to observe IvE in practice:

> *"So that at least the teachers on campus can, you know, come and observe. And get a better understanding of what invention learning could look like*." [Participant 2, Professor, CA]

Participant 3, a high school teacher in CT, similarly suggested that teachers could be convinced by seeing how students respond to creative and inventive learning:

> *"My convincing to them would be, like, look at the engagement of your students when you're doing something that you know has a little inventive, creative part to it, versus something that's more rote, or something that's a little bit more vanilla*."[Participant 3, High School Teacher, CT]

Beyond gaining buy-in, participants emphasized that IvE requires access to people with specialized knowledge. Participant 1 explained:

> *"Whatever problem the kids have, whatever area it is, you need an expert in that area*." [Participant 1, High School Teacher, CA]

Across these excerpts, the challenge is not simply recruiting additional staff. Participants described a need to develop shared understanding among teachers and administrators while also building networks of mentors and experts capable of supporting authentic student inventions.

Training-Related Challenges covers barriers associated with preparing educators and other staff to implement IvE effectively and providing continued support after their first exposure. Participants discussed differences in training quality, difficulties coordinating professional development, limited follow-up, insufficient administrative support, and the challenge of keeping staff knowledge updated as technologies and program expectations change.

In this subcategory, some participants highlighted the current lack of trained teachers. There weren't enough trained Teachers and educators to teach the IvE classes offerings, whether semester-long classes or electives. The Most excerpts by many participants explored this lack of trained teachers and educators.

A few participants did lay out the other facets of this training-related barrier for IvE. Participant 5, industry partner in CA, mentioned the one facet: considerable variation in the preparation of volunteers and staff:

> *"Some of them [have] varying degrees of quality. Some of them are really amazing, some of them are not as good with kids. You know, it's just because they're volunteers."*[Participant 5, Industry Partner, CA]

This illustrates that receiving training does not necessarily ensure that staff members are equally prepared to engage students or facilitate IvE activities. Participants also emphasized that training must be sustained rather than delivered as a one-time experience. Participant 11 explained:

> *"It's difficult to find time where we can meet with everybody, the teachers. So I think a more sustained, like, through the year, training sessions would be really helpful."* [Participant 11, University Professor, CA]

The problem continues after educators complete training. Participant 2 described the lack of a clear pathway for continued implementation:

> *"I get their buy-in and excitement, and then it's like they leave my class, and it's now what*?"[Participant 2, University Professor, CA]

Participants also noted that trained teachers may return to schools where administrators do not support IvE. As Participant 2 stated:

> *"They might not be supported by an administrator who supports invention education."[*Participant 2, University Professor, CA]

Technology creates an additional training burden because tools and equipment continually change. Participant 23 noted:

> *"We have a lot of technology, but the issue that a lot of our teachers face is technology has become outdated. It's not working or a part is missing*." [Participant 23, High School Teacher, TN]

Together, these excerpts suggest that training must be consistent, practical, ongoing, and supported by school leadership. Initial enthusiasm may not translate into sustained implementation when educators lack follow-up, time for continued professional development, or assistance adapting to new technologies.

Teacher Capacities encompasses barriers related to educators' confidence, skills, time, motivation, and willingness to adopt invention-based approaches. Participants emphasized that IvE often requires teachers to move away from familiar instructional practices, facilitate open-ended projects, take risks, and work across disciplinary boundaries.

For some educators, the transition to IvE can be intimidating because it differs from the teaching methods they have used for many years. Participant 22 explained:

> *"I don't think that it's an easy task for, like, the typical teacher that did whatever they did for that many years… to step into this new way of teaching."[Participant 22, Museum Director, CA]*

Risk-taking has emerged as an important part of teacher capacity. Teachers aren't taking enough risks for IvE program growth and impact. Participant 6 stated:

> *"Teachers need to be risk takers at times."[Participant 6, High School Teacher, MA]*

The same participant acknowledged that innovation may involve uncertainty or failure:

> *"There would be some failure involved, but ultimately, it would prove to be… a winner. So, you have to kind of take chances*."[Participant 6, High School Teacher, MA]

Participant 23 similarly described the challenge of presenting students with open-ended problems:

> *"Posing an open-ended problem for students to solve can be really intimidating for teachers, and takes a lot of time for teachers to wrap their heads around and then create the necessary supports."* [Participant 23, High School Teacher, TN]

Teacher capacity therefore involves more than technical knowledge. It includes confidence in facilitating open-ended learning, willingness to depart from traditional practices, comfort with uncertainty, and sufficient time to plan and support complex student projects. Even educators who value IvE may struggle to implement it when they feel unprepared, overextended, or unsupported.

School Logistical Constraints. The School Logistical Constraints subcategory describes barriers for implementing IvE in schools due to operational limitations that were not resource constraints or staffing barriers. This subcategory was divided into three sections, including *Time Constraints, Continuity Across Grade Levels*, and *Conflicting School Priorities.*

A major barrier detailed in this section was time-based constraints within the school day. Many educators mentioned that existing school day schedules include limited time for IvE enrichment. Rigid and packed school days led educators to foresee less opportunity to insert IvE into students' learning. For example, Participant 23, who is a High School Teacher in Tennessee describes their experience:

> *"Our middle schools are facing crunch time. Their time is so packed with all of these requirements. Now, more than ever, our middle schools are required to offer these high school courses, and all students must earn these certain high school courses, so their really feeling, like, packed"* [Participant 23, High School Teacher, TN]

Often, the lack of time in the school schedule for IvE was linked to school priorities. Traditional topics for education, including math, science, history, and English, were often seen as priorities compared to IvE integration. Participant 10, an Elementary School Leader from Tennessee, describes their experience working with state-based curriculum:

> *"It's a push and pull, a constant push and pull. Having been in another state before this, it was similar. Like, we're all fighting the standards movement and trying to find the happy medium of what's really … developmentally appropriate" [Participant 10, Elementary School Leader, TN]*

In addition to traditional subjects being prioritized for in-class time, some schools also prioritized test scores. Several educators commented on how pressure on schools and teachers to improve test results meant school days focused on test improvement practice. Test scores were also often seen as the only form of evaluation used to measure a student's success, leading to additional pressure to score well. High School Administrator from Tennessee, Participant 19, explains the issue as:

> *"We're gonna go in and talk about invention education, but there's gonna be people that say, hold on, I gotta talk about Algebra 1 scores, I gotta talk about, sophomore English scores, right? Because that's the only metrics that we currently have*" [Participant 19, High School Leader, TN]

Even when educators had the opportunity to see students for IvE based curriculum, there was also concern about the amount of time allotted. Educators mentioned a lack of frequency in

meetings with students. One High School Teacher from Connecticut described the issue as:

> *"The biggest limitation is that we don't meet frequently, and so as much as the students would want to be doing more with that. We got 90 minutes every week or so, and that's how much time we get to put into it, and we … proceed forward incrementally, but slowly"* [Participant 3, High School Teacher, CT]

Another barrier was a lack of continuity in curriculum across grades. It was commented in several interviews that there were specific gaps in middle school education that led certain middle schools to fall behind others. At a couple of schools interviewed, the schools had discrepancies in their IB programs. Participant 1, a High School Teacher from California, describes these challenges:

> *"So the school district should have done both middle schools, but they only did one. So half the kids have the IB experience coming into our school, and half of them don't. And so I think this is a disadvantage, right? Like, it would be better if both schools had design"* [Participant 1, High School Teacher, CA]

These challenges with curriculum continuity across middle school led certain students to being either better or worse prepared for their entrance to high school. While only a couple of schools reported this issue, differences in curriculum continuity both within a school district and across school districts will prepare students differently for entrance to high school.

Key Trends for Barriers by Region, Role, & Grade Level

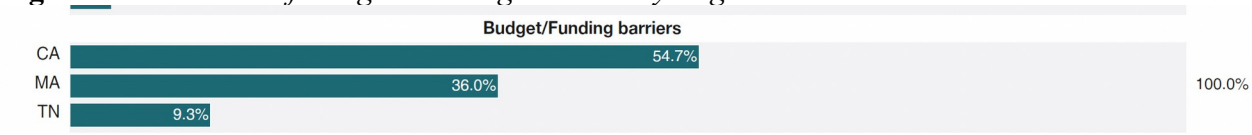
Examining the number of times resources are mentioned as a barrier and breaking up the codes by region reveals something interesting. California accounted for 62.9% of the normalized mentions of lack of resources, compared with 25.6% for Massachusetts and 11.5% for Tennessee (see Figure 22). This indicates that resource limitations were substantially more prominent in the California interviews than in the other two regions.

Figure 23. *Mentions of Lack of Resources by Region*

Normalized excerpt count for the selected code, grouped by descriptor.

Budget and funding concerns were also concentrated in California, which accounted for 54.7% of mentions, followed by Massachusetts at 36.0% and Tennessee at 9.3% (see Figure 24).

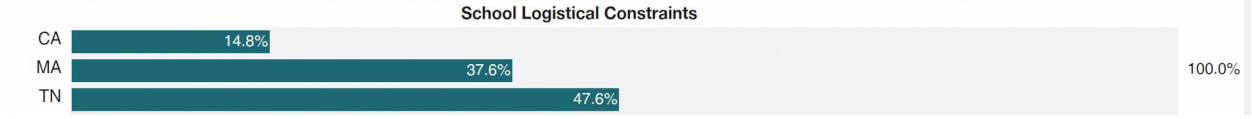
Figure 24. *Mentions of Budget/Funding Barriers by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

Tennessee accounted for the largest percentage of school logistical constraints at 47.6%, followed by Massachusetts at 37.6% and California at 14.8% (see Figure 25).

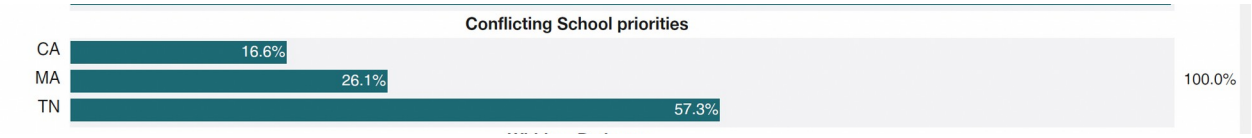
Figure 25. *Mentions of School Logistical Constraints by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

Conflicts between IvE and school priorities were most prominent in Tennessee, which accounted for 57.3% of mentions, compared with 26.1% in Massachusetts and 16.6% in California (see Figure 26).

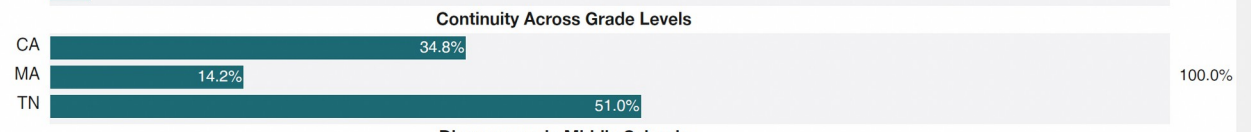
Figure 26. *Mentions of Conflicting School Priorities by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

Concerns about continuity across grade levels were most frequently associated with Tennessee at 51.0%, followed by California at 34.8% and Massachusetts at 14.2% (see Figure 27).

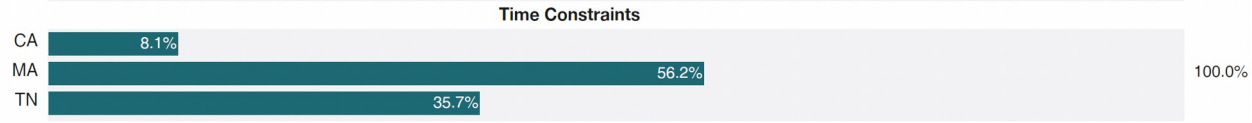
Figure 27. *Mentions of Continuity Across Grae Levels by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

Massachusetts accounted for the largest share of time-constraint mentions at 56.2%, followed by Tennessee at 35.7% and California at 8.1% (see Figure 28).

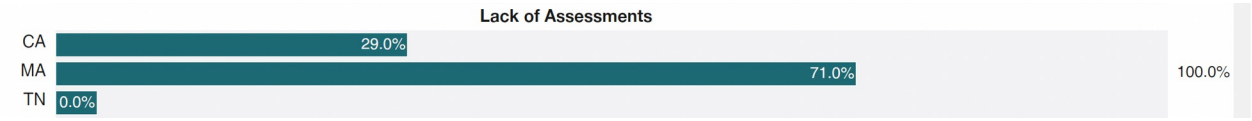
Figure 28. *Mentions of Time Constraints by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

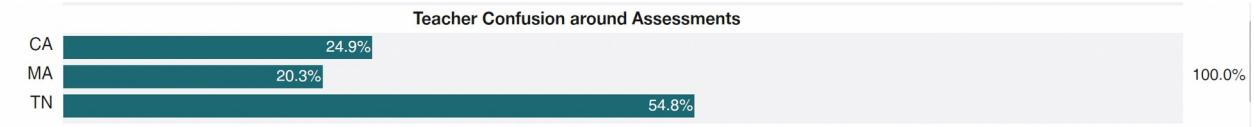
A lack of assessment was most prominent in Massachusetts, which accounted for 71.0% of mentions, compared with 29.0% in California and no mentions in Tennessee. In contrast, teacher confusion about assessment was most prominent in Tennessee at 54.8%, followed by California at 24.9% and Massachusetts at 20.3% (see Figures 29, 30).

Figure 29. *Mentions of Lack of Assessments by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

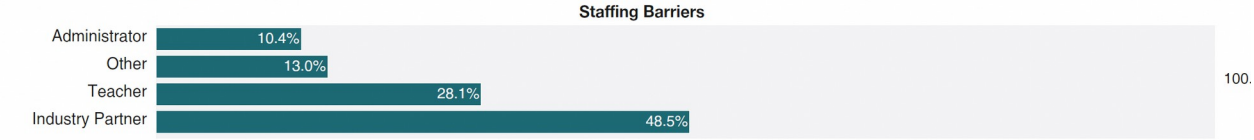
Figure 30. *Mentions of Teacher Confusion Around Assessments by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

Industry partners stood out among the interviewees when it comes to staffing barriers. 48.5% of excerpts pertaining to staffing barriers came from industry partners, while the next highest category, teachers, accounted for 28.1% of quotes (see Figure 31).

Figure 31. *Mentions of Staffing Barriers by Role*

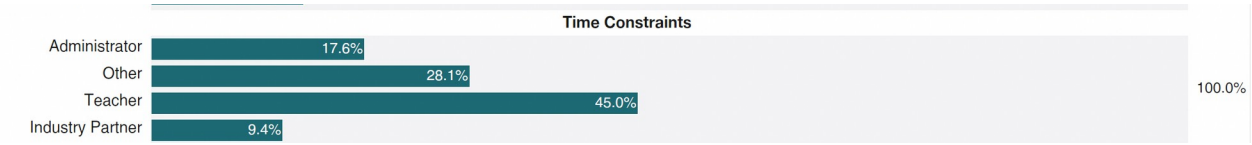


Normalized excerpt count for the selected code, grouped by descriptor.

Unsurprisingly, teachers reported the highest rate of issues with in-class time constraints. There is a significant discrepancy between teachers and administrators when it comes to this category

(45% vs. 17.6%), which could be a point of interest to examine further (see Figure 32).

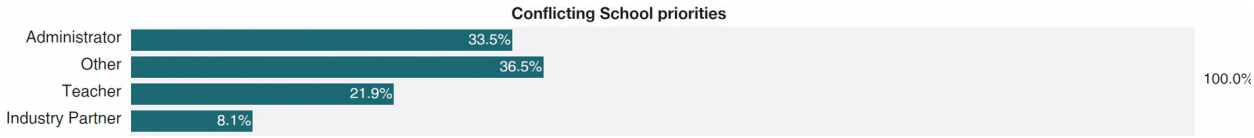
Figure 32. *Mentions of Time Constraints by Role*



Normalized excerpt count for the selected code, grouped by descriptor.

Administration and Other categories both talked more about school priorities conflicting with IvE than teachers did (see Figure 33). This also makes sense because the job of school leadership is to look at the goals of the entire school, while teachers focus more on classrooms.

Figure 33. *Mentions of Conflicting School Priorities by Role*



Normalized excerpt count for the selected code, grouped by descriptor.

Middle and high schools had the most issues with time constraints, at 35.7% each (see Figure 34). This could be in part because of the rigid schedule middle and high schools must follow. middle and high schools have to conform to tighter standards than elementary schools and also have a more limited school day than colleges.

Figure 34. *Mentions of Time Constraints by Grade Level*

Normalized excerpt count for the selected code, grouped by descriptor.

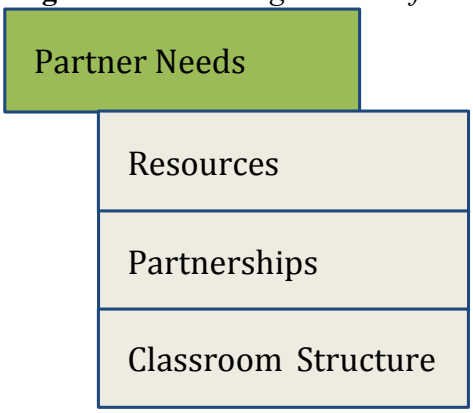
Overall, the findings indicate that barriers to IvE implementation are multidimensional and mutually reinforcing. Funding limitations affect not only access to materials and facilities but also teachers' compensation, training, and capacity to offer IvE opportunities. At the same time,

limited instructional time, competing academic priorities, inconsistent curricular pathways, and uncertainty about assessment make it difficult to integrate IvE into established educational structures. Differences across regions, roles, and grade levels further suggest that implementation strategies should be responsive to local contexts while addressing several shared needs: sustained funding, ongoing professional development, administrative support, stronger assessment approaches, and protected time for teachers and students to engage in IvE.

Overall Findings for Partner Needs

Partner Needs refers to what LMIT's regional partners need to continue offering, expand, or improve IvE. We grouped *Partner Needs* into three main categories: *Resources, Partnership*, and *Classroom Structure* (see Figure 35).

Figure 35. *Code Organization for Future Goals*



Resources. The resources category describes the different types of resources participants have stated needing in order to effectively expand IvE. This category is broken up into five subcategories: *funding, methods to educate others on IvE, updates/fixes in E-Labs, administrative support,* and *teacher trainings*.

One topic in particular that came up multiple times was the need for resources to inform others on what IvE is and how it can be implemented in the classroom. Participants expressed a need for a guide or online library where they can access information on IvE. Participant 15, an elementary school teacher from CA, shared:

> "*So having a location Where teachers can access. the lesson materials. and all of the… like, all of the attachments and accompanying stuff, I think would be the most helpful, because the easier it is to to access and to get into it, the more willing teachers are going to be able to do it"* [Participant 15, Elementary School teacher, CA]

Another need participants stressed was methods to train teachers in IvE. Many participants talked about having a lack of professional development in their school districts and regions. Some participants mentioned that participating in an invention sprint helped educators learn how to implement IvE because the educators themselves were participating in it. Participant 15 further stated:

> *"PD that LMIT has put on. This is my second year coming to one of the regional meetings, and I love the way we jump in with the invention sprint and just have teachers take on the role of inventor and solve that problem in like a quick way. I think that's really beneficial for teachers to just experience it in action"* [Participant 15, Elementary School teacher, CA]

Finally, participants expressed a need for more funding to help their programs. Funding is needed for materials, machines, professional development, and teacher salary. Funding is a major barrier that prevents schools from having larger and more effective IvE programs. Participant 18, a university administrator from MA, stated:

> *"Yeah, that kind of support helps me if I can get…support… for maybe reduced funding or some kind of a stipend, um, for the faculty to do the work… There are a lot of moving pieces to get people to actually engage in the implementation of what Lemelson MIT is putting out"* [Participant 18, University Administrator, MA]

Partnerships. Building and maintaining partnerships was one of the largest needs participants expressed when it came to expanding IvE. The areas that were most important to participants were building channels for communication between teachers and observing other schools implementing IvE. To help facilitate these conversations, several participants called on LMIT to help setup these partnerships.

One priority participants raised was ensuring that teachers themselves are directly involved in shaping IvE programs. Collaborating closely with teachers not only helps build something for a specific classroom but also creates resources that can extend beyond a single regional partnership. Participant 5, an industry partner from CA, shared:

> *"…working collaboratively with the teachers to create a program that fits their needs for invention education, their standards, but that could then also be given to teachers outside of this regional partnership with LMIT. I think that, that kind of working with the teachers is, a number one essential thing that we need."* [Participant 5, Industry Partner, CA]

Participants also stressed the value of connecting with educators in other regions who have experience implementing IV. Having established contacts to reach out to when facing challenges

was described as a critical resource especially when a curriculum isn't working as intended. Participant 3, a high school teacher from Connecticut, shared:

> *"Communication is important too. I know that there are some really experienced folks in Massachusetts that talking to them just for a little bit of time lets me know like, wow, okay, so this is the person you want to call and talk to when you're hitting a wall, when things aren't working, when this isn't fitting, when this doesn't make* sense." [Participant 3, High School Teacher, CT]

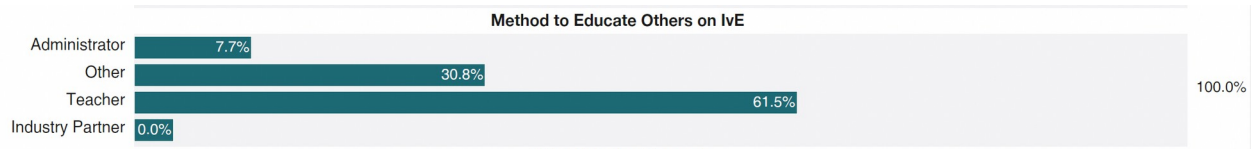
Finally, participants highlighted the importance of scoping out potential partnerships before committing to them. Having access to information and data about a region or community in advance helps schools understand local needs and make more informed decisions about their programs. Participant 7, a school principal from CA, stated:

> *"Being able to come in and look at those potential partnerships or those, you know, the data, like, what data is on our area to give us a head start on, like, what… what could we be looking at as we're… we're building these student groups to look at some of the… the needs that are in the different communities that we serve"* [Participant 7, School Administrator, CA]

Key Trends for Partner Needs by Role, Region & Grade Level

Key Trends by Role. There were several key trends that emerged when analyzing the frequency of codes for Partner Needs by Role, Region, and Grade Level. Teachers most strongly emphasized the practical challenge of helping others understand and adopt IvE. Teachers produced 61.5% of the excerpts coded as *Method to Educate Others on IvE*, compared with 30.8% from Other participants and 7.7% from administrators. They contributed the majority of discussions surrounding the need for resources that explain IvE to colleagues, students, families, and school leaders (see Figure 36). Rather than asking primarily for additional training themselves, teachers appeared to be concerned with becoming effective communicators of the approach. This reflects their position at the center of implementation, where they are responsible not only for teaching invention education but also for building understanding and support among those who are unfamiliar with it. Together, these findings suggest a complementary relationship between stakeholder groups: support personnel focused on preparing teachers, while teachers focused on equipping themselves to prepare others.

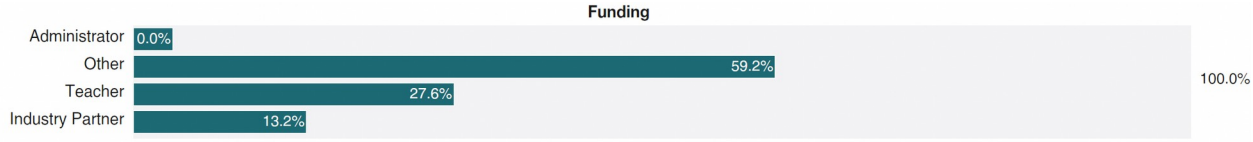
Figure 36. *Mentions of Method to Educate Others on IvE by Primary Role*



Normalized excerpt count for the selected code, grouped by descriptor.

Resource priorities also differed. Funding was concentrated among Other participants (59.2%), while teachers contributed 27.6% and industry partners 13.2% (see Figure 37). Discussions about financial resources were concentrated among participants outside the classroom, particularly those involved in coordination and higher education. Participants involved in coordination or higher education may have had greater visibility into grants, staffing, materials, and long-term program sustainability than classroom teachers. Classroom teachers, by comparison, rarely discussed financial sustainability, instead concentrating on the immediate resources needed to deliver invention education effectively.

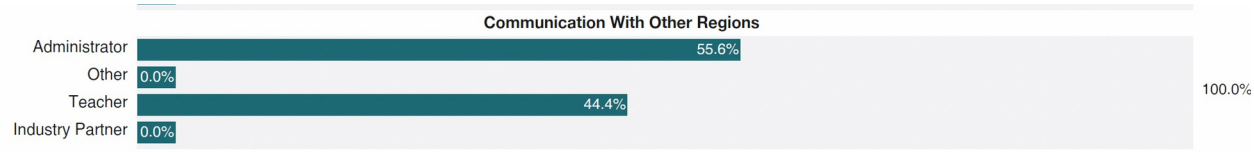
Figure 37. *Mentions of Funding Needs by Primary Role*



Normalized excerpt count for the selected code, grouped by descriptor.

Communication With Other Regions showed the reverse pattern: administrators contributed 55.6% and teachers 44.4%, while industry partners contributed none (see Figure 38). This indicates that educators and system leaders showed more interest during their interviews in learning from other regions' mistakes and innovations.

Figure 38. *Mentions of Communication with Other Regions by Primary Role*

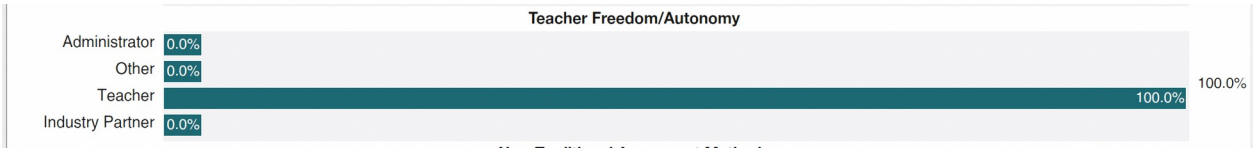


Normalized excerpt count for the selected code, grouped by descriptor.

Within Classroom Structure, teachers were the only role to mention Teacher Freedom/Autonomy (100%) (see Figure 39). This is an important contrast with the calls for training and consistency. Teachers did not simply request more prescriptive materials; they also argued that

implementation requires room to adapt projects, pacing, and classroom decisions to local students.

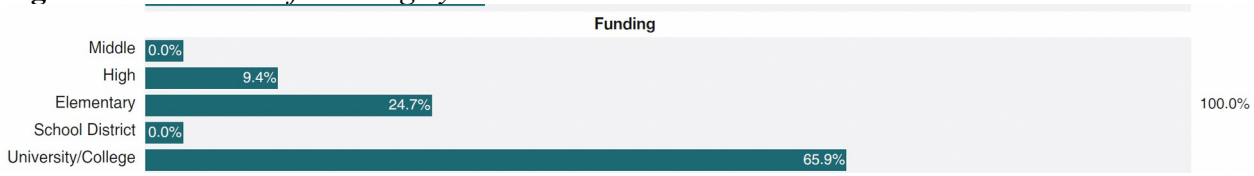
Figure 39. *Mentions of Teacher Freedom/Autonomy by Primary Role*



Normalized excerpt count for the selected code, grouped by descriptor.

Key Trends by Grade Level. When looking at the needs of regional partners, an interesting trend emerged with funding needs. The university/college level mentioned this the most by far, with 65.9% (see Figure 40).

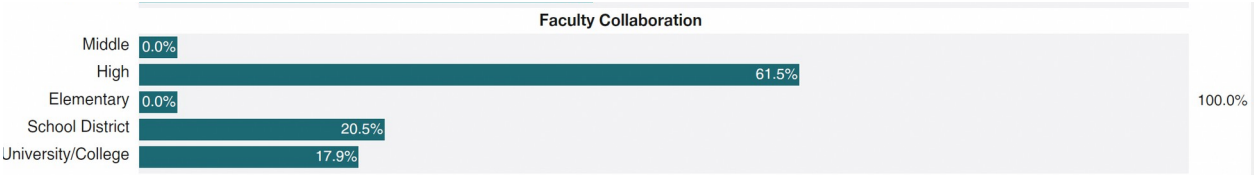
Figure 40. *Mentions of Funding by Grade Level*



Normalized excerpt count for the selected code, grouped by descriptor.

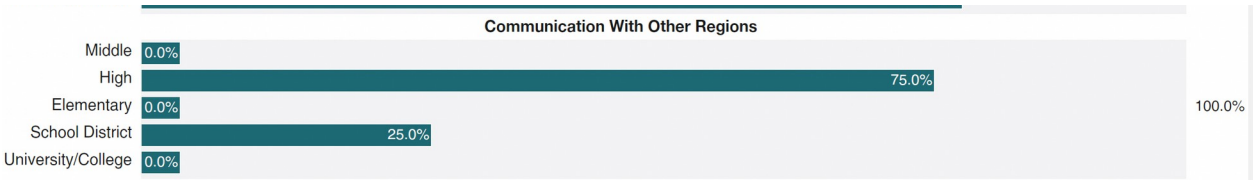
High schools also led the topics of faculty collaboration and communication with other regions, with 61.5% and 75%, respectively (see Figures 41, 42). Typical high schools have multiple educators teaching the same class, which could lead to an increased need for collaboration among faculty and communication with other regions.

Figure 41. *Mentions of Faculty Collaboration by Grade Level*



Normalized excerpt count for the selected code, grouped by descriptor.

Figure 42. *Mentions of Communication with Other Regions by Grade Level*

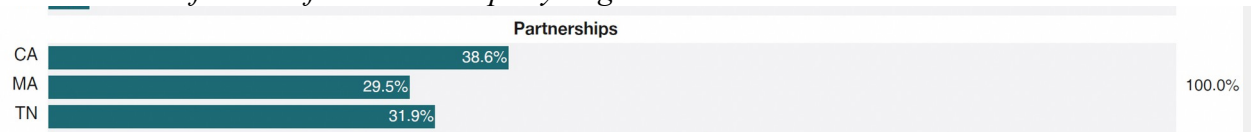


Normalized excerpt count for the selected code, grouped by descriptor.

Key Trends for Partner Needs by Region. The regional patterns differed across partnership, teacher-training, and funding needs.

Partnership-related needs were distributed relatively evenly across the three regions. California accounted for the largest share of mentions at 38.6%, followed by Tennessee at 31.9% and Massachusetts at 29.5% (see Figure 43). Although California had the highest percentage, the relatively small differences suggest that developing partnerships was a shared need across all three regions.

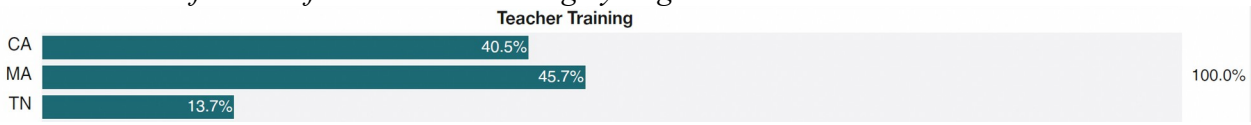
Figure 43. *Mentions of a Need for Partnerships by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

Teacher-training needs were concentrated primarily in Massachusetts and California. Massachusetts accounted for 45.7% of mentions, followed closely by California at 40.5%. Tennessee accounted for 13.7%, indicating that teacher training appeared less frequently in the Tennessee excerpts included in this analysis (see Figure 44).

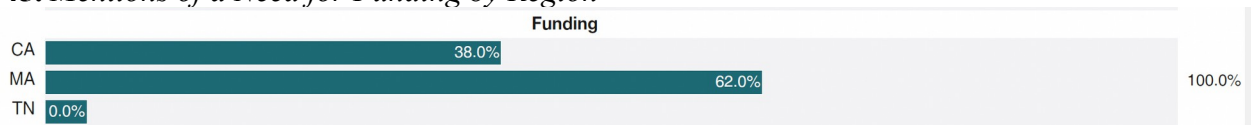
Figure 44. *Mentions of a Need for Teacher Training by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

Funding showed the greatest regional difference. Massachusetts accounted for 62.0% of funding-related mentions, while California accounted for 38.0%. No funding-related mentions were associated with Tennessee (see Figure 45). Among the three needs examined, funding was therefore the most geographically concentrated, with Massachusetts representing nearly two-thirds of the coded mentions.

Figure 45. *Mentions of a Need for Funding by Region*

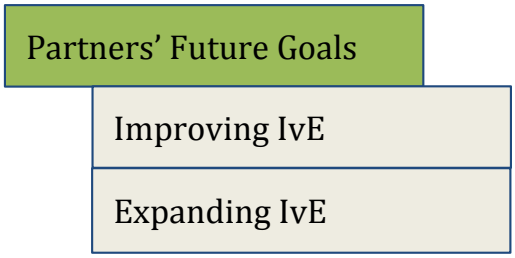


Normalized excerpt count for the selected code, grouped by descriptor.

Future Goals

The codes grouped under *Partners' Future Goals* represented what participants wanted the program to look like in the future. *Partners' Future Goals* have been broken down two major categories of *Improving IvE* and *Expanding IvE* (see Figure 46).

Figure 46. *Code Organization for Future Goals*



Improving IvE. Participants walked the interviewers through the future goals they hoped to accomplish in terms of improving IvE. The topics that emerged included *customizing IvE curricula, responsible AI utilization,* and *tracking students' progression through IvE courses across grade levels.*

One area of focus is looking at redefining curricula to ensure it matches the learning objectives for each grade level. Participants expressed interest in actively evaluating whether projects were the correct fit for their students rather than blindly following a curriculum. Participant 17, for example, mentions:

> *"Improving our pathway is just really looking at the curriculums and making sure that each grade is lined up with the right curriculum. We felt like toy design, it worked in second grade, but it wasn't really the best fit for them*." [Participant 17, Elementary School Leader, MA]

Along with curriculum design by grade level, basing more IvE curriculum around local industry was emphasized by Participant 18:

> *"Bringing in opportunities for projects that are real world projects, that our colleagues and industry partners and corporations and other, you know, big companies are bringing to our door for our students and our faculty to engage with*." [Participant 18, University Administrator, MA]

As AI tools become more accessible, participants also emphasized the importance of being intentional about how they are integrated into their IvE courses. Professional development can help staff and students understand how to use AI.

> *"So, we're very much AI-friendly… but we're very much intentional about how we kind of deploy it… Our students and staff are not going ballistic with Gemini, or ChatGPT, or whatever. They're very much conscientious about how they do wish to use it, and I think the training, the [professional development] really helped them realize it's not always going to be, you know, an answer to everything*." [Participant 21, University Administrator, CA]

Finally, several participants expressed interest in tracking student progression through IvE courses over time particularly at the elementary level. This continuity would offer an opportunity to observe how students grow their skills over the course of the program.

> *"And I think it will be really helpful because at the elementary level, the teacher will be seeing the same students as they move through the grades, so they'll really be able to note that progression."* [Participant 4, School District Administrator, MA]

Expanding IvE. When asked about future goals surrounding IvE one of the emergent themes was the expansion of IvE offerings. This expansion often came in the context of either expansion *within a school* or expansion *within the broader community*.

When talking to school administrators and teachers about growing the IvE offering the currently offered one of the most common ways they planned on doing that was creating a standalone course which could focus just on IvE curricula. It seems that only having IvE in shorter courses or as an adjunct to traditional courses limited the impact that it could have on students. So the hope is that transitioning to a longer, more focused IvE course will give schools more of an opportunity to fully immerse students in IvE and give students more fruitful results. Participant 12 a high school teacher from the state of Connecticut states:

> *"Our goal is to have a full semester-long course built. In addition to that, we're going to try to embed other, not full classes, but at least larger units or larger multi-day, multi-week invention-themed projects into some of our existing courses*." [Participant 12, High School Teacher, CN]

Our partners similarly expressed interest in expanding the IvE curriculum across grades to encompass the K-12 pathway. It seems the hope is that students are fully immersed and comfortable with IvE throughout their entire education. Students being familiar with IvE from

previous grades would also make implementation easier for subsequent grades according to some of the partners. Participant 23 states it clearly:

> *"...my vision is that students will have had the opportunity to invent and be creative from the moment that they step into our schools as a pre-K or a kindergarten student. all the way to 12th grade"* [Participant 23, High School Teacher, TN]

The discussion about expansion in the context of the broader community often revolved around the idea of expanding the amount of partnerships that a given school had. More partnerships gives access to more resources and more opportunities for students. These partnerships could be both with local businesses and other schools implementing IvE.

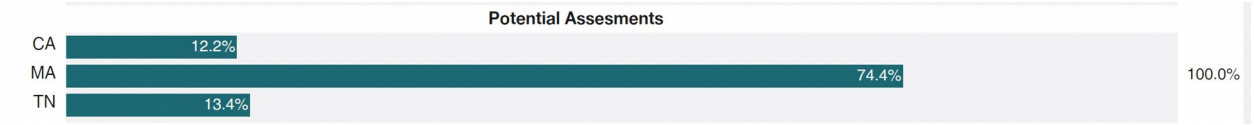
> *"I want to start to see how —. More so, community partners and community development is at the heart of, say, invention education, particularly when it comes to… Students starting to, regardless of their age or what year of schooling they are in, start to have access to community parks that can provide tangible issues that are impacting them on the daily basis."* [Participant 24, University Professor, CA]

Key Trends for Future Goals by Region, Role, and Grade Level

Key Trends by Region. Within Future Goals, a few key trends emerged when analyzing interview data by region. *Improving IvE* was captured mostly by excerpts from participants in the Massachusetts region, with an especially high concentration on possible assessment options.

The Massachusetts region was the most focused on *Future Goals* as a whole, contributing to 44.5% of all normalized excerpts. Of the responses illustrated by Figure 47, multiple Massachusetts excerpts mentioned the Future 9 Competency Assessment as an implementation goal, while the remaining excerpts focused on a general need for IvE assessments for better longitudinal metrics.

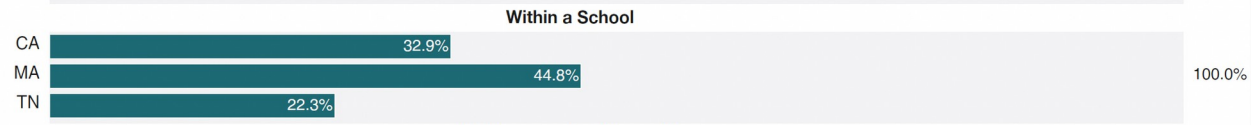
Figure 47. *Potential Assessment by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

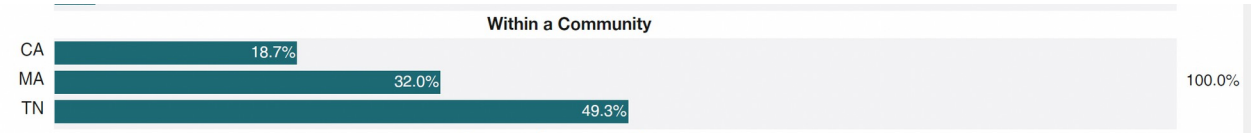
A secondary Future Goal captured by participants in Massachusetts is highlighted, but less prominently. Massachusetts focused the most on *Expanding IvE* within a school out of all the regions (see Figure 48), while Tennessee participants focused more on *Expanding IvE* within the community (see Figure 49).

Figure 48. *Expanding IvE: Within a School by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

Figure 49. *Expanding IvE: Within a Community by Region*



Normalized excerpt count for the selected code, grouped by descriptor.

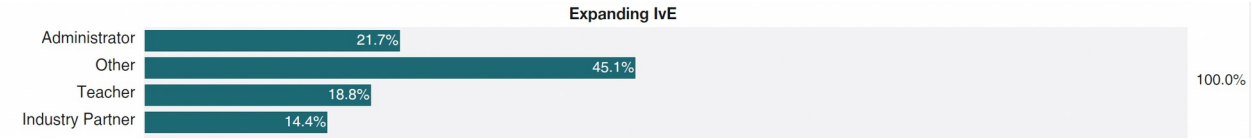
Both collections of excerpts mention a need for contextual implementation of IvE. Although, Tennessee excerpts focused more on partnerships and communication, and Massachusetts excerpts on adding IvE as a separate course and increasing access to the average student. This speaks to the capacities of both regions; The implementation of VW ELabs in Tennessee provides a springboard for IvE within individual schools, and their goals align more with expansion. Massachusetts on the other hand, a region which lacks a singular dominant partnership, is more focused on Future Goals at the school level. This presents the idea that creating a central root of IvE within a region, like a single strong partnership or established labs, may allow a region to orient their efforts more towards the broader community.

Key Trends by Role. The figures below provide a clear representation of the two topics in Future Goals. All roles mentioned expanding partnerships and K-14 pathways, yet Teacher excerpts focus more on training pre-service teachers on IvE; Excerpts from Administrators and Other highlight communication with other regions and getting students workplace experience.

The Administrator and Other roles combine to 74% of the normalized excerpts in Future Goals surrounding *Improving IvE*, while the focus of the Teacher role is 10% (see Figure [0] for normalized excerpt count). For *Expanding IvE*, Administrator and Other roles make up about 67% of the normalized excerpts, while the Teacher role jumps up to 18.8% (Figure [3]).

When examining the normalized excerpt counts for both figures, it became apparent that every single role mentioned *Expanding IvE* more than *Improving IvE (see Figures 50, 51)*. Combined with the higher proportion of Teacher excerpts captured in *Expanding IvE,* it may be the case that expanding offerings, such as adding more IvE courses and making a replicable IvE model is a higher priority than improving on existing curricula and assessments.

Figure 50. *Expanding IvE excerpts by Role*



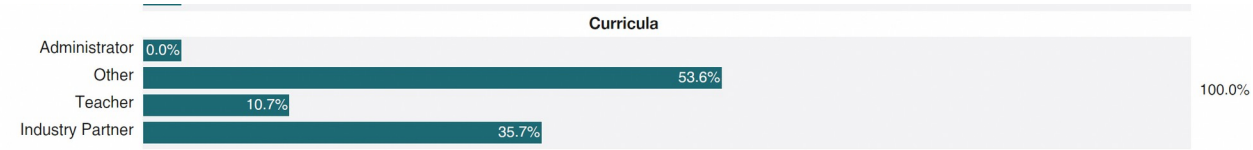
Normalized excerpt count for the selected code, grouped by descriptor.

Figure 51. *Improving IvE excerpts by Role*

Normalized excerpt count for the selected code, grouped by descriptor.

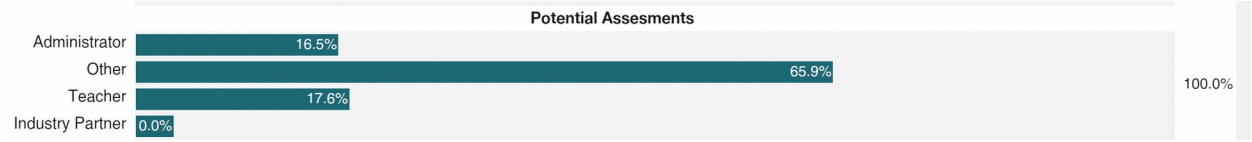
Figures 52 and 53 dive deeper into *Improving IvE,* showing that Administrators are more focused on improving IvE curricula, while the "Other" role (other types of educational leaders) is focused on improving potential assessment measures for IvE. The lack of responses from Industry Partners on assessment is likely due to many Industry Partners, like museums, providing short-term IvE activities that do not require long-term assessment metrics. As part of IvE improvement, administrators and industry partners also mentioned incorporating responsible AI use, at 69.2% and 30.8%, respectively (see Figure 54).

Figure 52. *Improving IvE: Curricula by Role*



Normalized excerpt count for the selected code, grouped by descriptor.

Figure 53. Improving IvE: Potential Assessment by Role



Normalized excerpt count for the selected code, grouped by descriptor.

Figure 54. Responsible AI use by role

Normalized excerpt count for the selected code, grouped by descriptor.

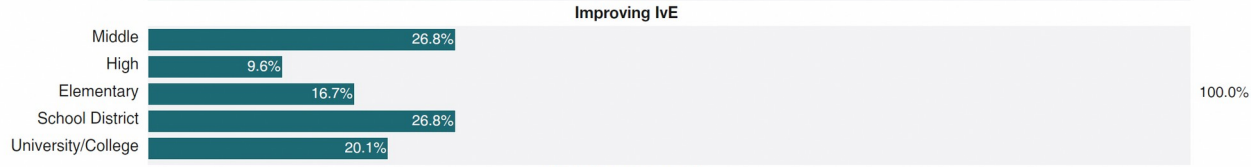
Overall, the Future Goals group, when examined by role, suggests that administrators and other participants in leadership positions articulated many goals related to *Improving IvE* and *Expanding IvE* within their organizations. The comparatively limited number of teacher responses in this group may reflect differences in interview emphasis, as teachers more frequently discussed barriers and capacity-related concerns. It is also possible that teachers' fewer comments about *Improving IvE* reflect the extent to which improvement efforts depend on broader organizational support and strong professional networks. Even so, the substantial proportion of Future Goals excerpts contributed by participants in leadership roles is encouraging, as it suggests that those in positions to influence implementation are thinking actively about the continued development and expansion of IvE in the coming years.

Key Trends for Future Goals by Grade Level

The Future Goals excerpts showed an interesting trend when broken down by Grade Level instructed by participants. The Middle School and School District levels had the highest proportion of excerpts captured about Improving IvE, at 26.8% each (see Figure 55).

Figure 56 echoes sentiments captured from many participants: The Middle School grade level is often the weakest part in a given K-14 pathway. Low student engagement and strict state standards being the most prominent reasons that emerged from participants. Figure 56 shows no responses from Middle School grade level regarding IvE Expansion. The High School grade level shows a trend in the opposite direction, with *Expanding IvE* garnering a larger proportion of excerpts than *Improving IvE*.

Figure 55. *Improving IvE by Grade Level*



Normalized excerpt count for the selected code, grouped by descriptor.

Figure 56. *Expanding IvE by Grade Level*

Normalized excerpt count for the selected code, grouped by descriptor.

Discussion

RQ1. How do regional partners conceptualize and implement Invention Education?

Partners described IvE primarily as a way to develop skills and habits of mind through open-ended inquiry. References to soft skills were much more frequent in the interview material (117 mentions) than references to technical skills (19 mentions) or shifts in student mindset and identity (30 mentions). These are counts of mentions, rather than counts of participants, but they suggest that partners see IvE's value extending beyond the technical work of making an invention. As one participant explained, an effective experience is "structured, but it's open-ended", with the open-ended element being "really critical" (Participant 4, School District Administrator, MA).

Partners' descriptions also suggest that IvE is being implemented through different instructional approaches, rather than a single prescribed curriculum. An elementary teacher, for example, described inquiry as a guiding principle: "we don't use, like, scripted curriculum. We design our own curriculum in different subject areas" (Participant 15, Elementary School Teacher, CA). This flexibility may help educators connect invention activities to local needs and existing courses. It also creates a need to identify the elements that make an experience recognizably IvE across settings.

Some partners connected IvE to a newer concern: students' use of AI tools. One high school administrator suggested that IvE could help students "think a little bit deeper" as AI changes how they learn (Participant 7, High School Administrator, CA). A teacher similarly saw

it as a possible response to "students handing their agency and their thinking off to AI" (Participant 3, High School Teacher, CT). These comments express partners' expectations, rather than evidence of an observed effect. They nevertheless point to student agency and independent problem solving as outcomes worth examining as pathways develop.

RQ2. What conditions influence the development and scaling of K–14 IvE pathways?

The interviews indicate that pathway development depends in part on whether teachers have room to adapt instruction and try unfamiliar approaches. Participants described risk-taking as important to IvE, while also recognizing that teachers' freedom varies across schools and districts. One experienced teacher worried that newer colleagues "don't feel like they have the ability to take a different path" (Participant 13, Elementary Teacher, MA). This suggests that professional development alone may be insufficient if local expectations leave little space for teachers to put what they learn into practice.

Partners also emphasized the need to make deliberate choices about how IvE fits a particular community. A participant proposed forming a committee to "map out this pathway" and decide "what are the outcomes we want to have?" (Participant 4, Science Director, MA). Another stressed that a school community cannot focus on every possible outcome and must decide "where are we going to put our focal point?" (Participant 19, High School Administrator, TN). Together, these comments suggest that scaling will require a shared pathway planning process with enough flexibility for regions to select priorities that fit their students, resources, and existing programs.

Participants expressed pride in established local programs, including school fab labs and strong science experiences. Those programs may provide a foundation for IvE pathways, but a pathway also needs a reliable place in students' schedules. Partners described both dedicated courses and substantial projects within existing courses as possible routes. One university professor outlined a goal of building a semester-long course while adding "larger multi-day, multi-week invention-themed projects" elsewhere in the curriculum (Participant 2, University Professor, CA). These approaches could expand access in different ways; documenting when and where students encounter IvE will be important for understanding whether experiences form a continuous pathway across grade spans.

RQ3. What role should LMIT play in supporting these efforts?

Partners identified a need for language that helps them explain IvE to teachers, administrators, families, and potential collaborators. One administrator cautioned that discussions can become "educational jargonese" and asked how partners could simplify their message (Participant 19, High School Administrator, TN). A clear, communicable definition could give regions a shared account of IvE's purpose while leaving room for different local activities and instructional models.

Partners also saw value in learning from one another. A professor pointed to Chattanooga's sustained use of project-based learning across grade levels as an example worth considering (Participant 11, Professor, CA), while another partner asked for "thought partners" from both LMIT and the other regions (Participant 25, High School Administrator, TN). Regular exchanges could help partners compare pathway designs, discuss implementation problems, and adapt ideas that fit their own settings.

LMIT may also be able to help regions identify and engage potential partners. A university administrator described LMIT as a prospective "true convener in the region" and said the association could lend credibility to local outreach (Participant 18, University Administrator, MA). Another participant asked for help examining regional data and potential partnerships so students could investigate needs in the communities they serve (Participant 7, High School Administrator, CA). These requests suggest a practical convening role such as helping partners connect organizations and local problems to meaningful student invention work.

Finally, participants requested curriculum examples that are easy to find, use, and adapt. Needs differed by setting: an elementary teacher wanted a clear location for lessons and accompanying materials (Participant 15, Elementary Teacher, CA), a museum director wanted a ready-to-run 30-minute activity (Participant 22, Museum Director, CA), and a university administrator sought materials that could be brought to the college level (Participant 18, University Administrator, MA). Full-year examples by grade level could offer starting points for new teachers, while shorter modules and adaptable materials could serve experienced educators and informal learning spaces. LMIT could use partner COP convenings to gather feedback on these resources and improve their fit across contexts.

Conclusion

Across Northern California, Tennessee, and Massachusetts, partners conceptualized IvE as a flexible, authentic approach to learning that develops technical competencies alongside creativity, collaboration, problem solving, and students' identities as capable inventors and problem solvers. At the same time, differences in organizational capacity, available resources, partnerships, educator preparation, and regional priorities shape how IvE is currently implemented and the opportunities and challenges partners encounter in expanding it. The findings suggest that building sustainable PreK-14 IvE pathways requires more than introducing individual programs or curricula; it requires coordinated regional infrastructures that connect educators, schools, higher education, industry, community organizations, and other partners around a shared vision for invention learning.

As LMIT moves from regional exploration toward pathway development, implementation, and scaling, these findings provide an important foundation for determining where and how to invest resources and support. Continued attention to educator capacity building, cross-sector collaboration, curriculum and assessment alignment, sustainable funding and coordination, and mechanisms for sharing knowledge across regions will be critical. Equally important will be continued research and evaluation that documents how regional pathways develop over time and examines their influence on educator practices and student outcomes.

References

Brinkman, S., & Kvale, S. (2015). Interviews: Learning the craft of qualitative research interviewing (3rd ed.). Sage.

Couch, S., Skukauskaitė, A., & Estabrooks, L. B. (2019). Invention education and the developing nature of high school students' construction of an "inventor" identity. Technology & Innovation, 20(3), 285-302. /https://doi.org/10.21300/20.3.2019.285

Coy, A. (2020). A framework for invention education. *Lemelson Foundation*.

Dalela, S., & Ahmed, M. S. (2024). Systematic review of invention education research landscape: State of the discipline and future directions. Frontiers in Education, 9, Article 1284442. https://doi.org/10.3389/feduc.2024.1284442

Davies, D., & Dodd, J. (2002). Qualitative research and the question of Rigor. Qualitative Health Research, 12(2), 279-289. https://doi.org/10.1177/104973230201200211

Garner, J., Matheny, E., Rutledge, A., & Kuhn, M. (2021). Invention education as a context for children's identity exploration. J. STEM Outreach 4, 1–14. doi: 10.15695/jstem/v4i1.07

National Academies of Sciences, Engineering, and Medicine (2024a). *Equity in K-12 STEM Education: Framing Decisions for the Future*. Washington, DC: The National Academies Press.

National Academies of Sciences, Engineering, and Medicine (2024a). *Equity in K-12 STEM Education: Framing Decisions for the Future*. Washington, DC: The National Academies Press.

National Academies of Sciences, Engineering, and Medicine (2024). Scaling and Sustaining Pre-K-12 STEM Education Innovations: Systemic Challenges, Systemic Responses. Washington, DC: The National Academies Press. https://doi.org/10.17226/27950.

Roulston, K. (2022). Interviewing: A guide to theory and practice. Sage.

Tracy, S. J. (2010). Qualitative quality: Eight "big-tent" criteria for excellent qualitative research. Qualitative Inquiry, 16(10), 837-851. https://doi.org/10.1177/1077800410383121

Appendix A: Interview Guide

Study Title: Advancing K-14 Invention Education Pathways: Insights from Regional Partners and Collaborators

Duration: 45 minutes

Location: Zoom

The interviewers briefly introduce themselves.

Let them know that they are being recorded and ask if they are ok with it.

Verbal Consent

You have been asked to participate in a research study conducted by the Lemelson-MIT (LMIT) program researchers. You were selected as a possible participant in this study because you are part of LMIT's regional collaborators' group whose views are being examined in relation to their current capacity, challenges, and supports needed from LMIT to implement and scale Invention Education (IvE) within and across regions.

The information below provides a summary of the research. Your participation in this research is voluntary and you can withdraw at any time.

- **Purpose**
  We want to understand our regional partners' current capacity, needs, and support needed to implement and scale IvE offerings in their organizations and regions.
- **Study Procedures**
  We ask that you complete a 45-minute interview via Zoom that will be recorded, transcribed, stored securely, analyzed, and reported.
- **Risks & Potential Discomfort**
  The only risks, discomforts, or inconveniences that we can foresee are the possibility that your real name or identity may become apparent even if a unique ID is used, and that some questions might make you feel uncomfortable. You may skip any question you do not wish to answer.
- **Alternatives to Participation**
  You may choose not to take part in the interview. Refusal to participate in the study will not impact your ability to collaborate with LMIT and participate in program offerings affiliated with LMIT.

Opening:

- Can you briefly introduce your role and your organization?
- How would you define Invention Education (IvE) in the context of your organization and region?

Current Capacity & Assets

Q1. What do you perceive as the opportunities for building K-14 IvE pathways in your organization and region?

a) What IvE-related programs, courses, or experiences are currently offered in your organization/region?
b) What existing partnerships support your IvE efforts? How do they function?
c) What other strengths or assets does your organization bring to IvE pathway building (e.g., staff expertise, facilities, partnerships, funding)?

Q2. What needs in your community will IvE help to address?

Student Learning Outcomes & Impact

Q4. What learning outcomes do you hope students gain from IvE experiences? (probe: What learning outcomes matter most to your organization (e.g., skills, career pathways)?

a) What quantitative or qualitative measures will help to demonstrate the outcomes?
b) How do you currently measure/assess student learning or success in IvE-related programs (if at all)? (Probe: What type of assessments /assessment frameworks are you currently using?)

Needs & Supports Required

Q5. What are the needs and types of support that would help your organization expand or strengthen IvE?

a. What professional development or training do educators need to effectively implement IvE?
b. What resources (curriculum, tools, assessment systems, technology) are most needed?
c. What kinds of partnerships or connections would strengthen your IvE efforts?
d. How can LMIT best support your organization or region?

Q6. What would a strong IvE pathway look like in your region (middle school - high school - college/career)?

Q7. Is there anything we didn't ask that you feel is important for understanding your needs or context?

Thank you for your participation!